plete if we note that $(\phi, g) = 0$ means that g is orthogonal to the solution of

$$K_s^\dagger \phi = \lambda^* \int k_s(y, x)\, \phi(y) dy = \phi$$

when $\lambda = 1/(\phi, \phi)$.

Thus there is a close relation between this simple infinite-dimensional example and the finite-dimensional case. More interesting, however, is the hint that the divergence of the Neumann series is somehow related to the occurrence of a solution to the homogeneous equation.

Because separable kernels lead to integral equations with extremely simple closed-form solutions, physicists have naturally attempted to make use of them in practical problems. One example of their utility is found in the study of nuclear forces. For many years physicists have tried to determine the form of nucleon-nucleon potentials, and in doing so were forced to discard practically every sacred classical concept of what a potential should be. As mentioned in Chapter 1, the requirement of central potentials had to be abandoned, and it also had to be admitted that nuclear forces might depend on the intrinsic spin of the nuclear particles. It was natural that eventually even the requirement of *locality* would come into doubt.

By a *local potential*, we mean one whose action at a point depends only on what is happening at that point. Thus Schrödinger's equation is traditionally written with a local potential:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}) \,. \tag{8.63}$$

However, it cannot be ruled out a priori that in quantum mechanics the potential may be an operator which acts in a nonlocal manner. For example, we could imagine writing Schrödinger's equation as

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \int U(\mathbf{r}, \mathbf{r}')\psi(\mathbf{r}')\, d\mathbf{r}' = E\psi(\mathbf{r}') \,. \tag{8.64}$$

We see from this equation that the effect of the potential operating on ψ does not depend just on the value of ψ at one point $\mathbf{r}$, as is the case in Eq. (8.63), but on the integral over some region of space weighted by the potential function $U(\mathbf{r}, \mathbf{r}')$. Note that if we want the Hamiltonian of Eq. (8.64) to be Hermitian, we must require that

$$U(\mathbf{r}, \mathbf{r}') = U(\mathbf{r}', \mathbf{r})^* \,, \tag{8.65}$$

The Green's function technique developed in Chapter 7 can readily be applied to Eq. (8.64). Proceeding in the manner described in Section 7.4, we find that $\psi(\mathbf{r})$ is given by the integral equation

$$\psi(\mathbf{r}) = \frac{A}{(2\pi)^{3/2}} e^{i\mathbf{q}_i \cdot \mathbf{r}} - \frac{m}{2\pi\hbar^2} \iint \frac{e^{iq_i|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|} U(\mathbf{r}', \mathbf{r}'')\psi(\mathbf{r}'')\, d\mathbf{r}'\, d\mathbf{r}'' \,, \tag{8.66}$$

where $\mathbf{q}_i$, m, $\hbar$, and A have the same significance as in Section 7.4. Note the similarity between this equation and Eq. (7.68). As was mentioned in Section

7.4, one of the most interesting physical quantities is the scattering amplitude $f^s(\mathbf{q}_f)$, which is defined in terms of the amplitude of the outgoing scattered wave. If we proceed as in Section 7.4, it is a simple matter to show that $f^s(\mathbf{q}_f)$ is given by

$$f^s(\mathbf{q}_f) = f(\mathbf{q}_f) , \tag{8.67a}$$

where f satisfies the integral equation

$$f(\mathbf{k}) = -\frac{4\pi^2 m}{\hbar^2}\hat{U}(\mathbf{k}, \mathbf{q}_i) - \frac{2m}{\hbar^2}\int \hat{U}(\mathbf{k}, \mathbf{q}) \frac{1}{q^2 - (q_i + i\epsilon)^2} f(\mathbf{q})\, d\mathbf{q} , \tag{8.67b}$$

where $\hat{U}(\mathbf{k}, \mathbf{q})$ is the Fourier transform of $U(\mathbf{r}, \mathbf{r}')$, that is,

$$\hat{U}(\mathbf{k}, \mathbf{q}) = \frac{1}{(2\pi)^3}\iint e^{i\mathbf{q}\cdot\mathbf{r}'} e^{i\mathbf{k}\cdot\mathbf{r}}\, \hat{U}(\mathbf{r}, \mathbf{r}')\, d\mathbf{r}\, d\mathbf{r}' .$$

In Eqs. (8.67a) and (8.67b), $\mathbf{q}_i$ represents the incident direction of the particle and $\mathbf{q}_f$ represents its final direction. The magnitudes of $\mathbf{q}_i$ and $\mathbf{q}_f$ are equal and given by

$$|\mathbf{q}_i| = |\mathbf{q}_f| = \left[\frac{2mE}{\hbar^2}\right]^{1/2} .$$

Once we have $f(\mathbf{k})$, all the quantities of interest in the scattering process can be determined. For a general $U(\mathbf{k}, \mathbf{q})$ there is not much we can say about this function $f(\mathbf{k})$, but if we take U to be of the *separable* form, that is,

$$U(\mathbf{r}, \mathbf{r}') = -g^2 v(\mathbf{r}) v(\mathbf{r}') , \tag{8.68}$$

then we can make progress. Here $v(\mathbf{r})$ is a real-valued function, in order to insure that U is Hermitian. Using this form for U and denoting the Fourier transform of $v(\mathbf{r})$ by $\hat{v}(\mathbf{k})$, we bring Eq. (8.67b) to the form

$$f(\mathbf{k}) = \frac{4\pi^2 m g^2}{h^2}\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k}) + \frac{2mg^2}{h^2}\int \hat{v}(\mathbf{k}) \frac{\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2} f(\mathbf{q})\, d\mathbf{q} ,$$

or

$$f(\mathbf{k}) = \pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k}) + \frac{\tau}{2\pi}\int \hat{v}(\mathbf{k}) \frac{\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2} f(\mathbf{q})\, d\mathbf{q} , \tag{8.69}$$

where we have set $4\pi m g^2/\hbar^2 \equiv \tau$. But Eq. (8.69) has precisely the form of Eq. (8.54) if we make the identifications

$$x \to \mathbf{k} , \qquad y \to \mathbf{q} , \qquad g(x) \to \pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k}) , \qquad \phi(x) \to \hat{v}(\mathbf{k}) ,$$
$$\psi^*(y) \to \frac{\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2} ,$$

and set $\lambda = \tau/2\pi$. Of course, the volume of integration is three-dimensional, but in Eq. (8.54) one need not be so literal-minded as to associate x and y with one-dimensional vectors and dy with a one-dimensional volume. The solution

to Eq. (8.69) is given immediately by Eq. (8.61). We find that

$$f(\mathbf{k}) = \pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k}) + \lambda \frac{\int \frac{\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2}[\pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{q})]\, d\mathbf{q}}{1 - \lambda \int \frac{\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2}\hat{v}(\mathbf{q})\, d\mathbf{q}}\hat{v}(\mathbf{k}) ,$$

which can be written as

$$\begin{aligned} f(\mathbf{k}) &= \pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k})[1 + \lambda I/(1 - \lambda I)] \\ &= \pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{k})/(1 - \lambda I) , \end{aligned}$$

where we define

$$I \equiv \int \frac{\hat{v}(\mathbf{q})\hat{v}(\mathbf{q})}{q^2 - (q_i + i\epsilon)^2}\, d\mathbf{q} .$$

According to the definition of $f^s(\mathbf{q}_f)$, we see that

$$f^s(\mathbf{q}_f) = \frac{\pi\tau\hat{v}(\mathbf{q}_i)\hat{v}(\mathbf{q}_f)}{1 - \tau I/2\pi} . \tag{8.70}$$

Thus given any function $v(\mathbf{r})$ [or $v(\mathbf{q})$] the problem is solved. One obvious choice for $v(\mathbf{r})$ is a simple Yukawa potential, $v(\mathbf{r}) = e^{-\mu r}/r$, with the corresponding Fourier transform given by Eq. (7.58′):

$$\hat{v}(\mathbf{q}) = \left(\frac{2}{\pi}\right)^{1/2} \frac{1}{q^2 + \mu^2} . \tag{8.71}$$

Since this depends only on q^2, we see that $f^s(\mathbf{q}_f)$ will depend only on the magnitude of $\mathbf{q}_f$ (and of $\mathbf{q}_i$). We can therefore consider f^s to be a function of E alone, and not of the scattering angle. For this choice of $\hat{v}(\mathbf{q})$, the integral I can be readily computed by the contour integral techniques of Chapter 6. One finds that

$$I = -\frac{2\pi}{\mu} \frac{1}{(q_i + i\mu)^2} , \tag{8.72}$$

where we have let ϵ go to zero after performing the integration. Putting Eqs. (8.71) and (8.72) into Eq. (8.70), we find that

$$f^s(E) = \frac{2\tau}{(q_i^2 + \mu^2)^2 + \frac{\tau}{\mu}(q_i - i\mu)^2} . \tag{8.73}$$

If we make the definitions

$$a_0 \equiv \frac{2}{\mu}\left(1 - \frac{\mu^3}{\tau}\right)^{-1} , \qquad r_0 \equiv \frac{1}{\mu}\left(1 + \frac{2\mu^3}{\tau}\right) ,$$

then

$$f^s(E) = \left[-\frac{1}{a_0} + \frac{1}{2} r_0 q_i^2 + \frac{1}{2\tau} q_i^4 - iq_i\right]^{-1} . \tag{8.74}$$

We see that f^s has no angular dependence; i.e., it represents pure s-wave ($l = 0$) scattering.

It can be shown that for any well-behaved short-range *local* potential, the low energy ($\mathbf{q}_i$ small) behavior of the s-wave scattering amplitude is

$$f^s = \left[-\frac{1}{a_0} + \frac{1}{2} r_0 q_i^2 + \cdots - iq_i \right]^{-1} ,$$

where a_0 and r_0 can be calculated from the specific form of the local potential. (See, for example, Blatt and Weisskopf, *Theoretical Nuclear Physics*.) This is known as the effective range form of f^s; a_0 and r_0 are called the *scattering length* and the *effective range* respectively. Thus we see that it is possible to find a non-local potential, which can be handled in closed form, whose associated scattering amplitude looks just like the low energy scattering amplitude for a local potential. We leave it as an exercise for the reader to show that the above potential can also give rise to a bound state (i.e., a negative eigenvalue of Schrödinger's equation) if g is sufficiently large.

Because of the possibility of accommodating a bound deuteron and also because it gives rise to a scattering amplitude of the effective range type, the separable potential of Eq. (8.68) (and its generalization to spin-dependent forces) has often been used to analyze the low energy properties of the nucleon–nucleon system. In recent years, potentials of this type have also been used in theoretical studies of three-particle systems, where their simple form is very useful in obtaining tractable expressions for three-particle scattering amplitudes and bound state wave functions.

8.6 GENERAL KERNELS OF FINITE RANK

Now that we have seen how the separable kernel leads to a closed-form solution to an integral equation, it is not difficult to generalize to kernels of *finite rank*. The only complication is that instead of being able to reduce the vector space equation to a single scalar equation as we did for separable kernels, for kernels of finite rank we will reduce the vector space equation to a *system* of scalar equations. These equations can be written in the form of a finite-dimensional *matrix* equation, which is why such kernels are said to be of finite rank (see Section 3.10).

We define a *kernel of finite rank* to be of the form

$$K_F f \equiv \sum_{n=1}^{N} \phi_n (\psi_n, f) , \tag{8.75}$$

where f is any vector in the Banach space, B, on which K_F acts. The inner product used here need not be that appropriate to a Hilbert space; all we mean by (ψ_n, f) is a *linear functional*, denoted symbolically by (ψ_n, f), which takes any $f \in B$ into a scalar. The set of all linear functionals on a Banach space, B, is also a Banach space, often referred to as the *dual space* of B. In the case of a Hilbert space, this space of linear functionals is isomorphic to the Hilbert

space itself. For the space L_p, the Banach space of linear functionals can be identified isomorphically with the space L_q, where $1/p + 1/q = 1$.

It can readily be shown (see Problem 8.21) that if $f \in L_p$, and $g \in L_q$ and if $1/p + 1/q = 1$, then

$$(g, f) = \int g^* f \, dx$$

exists, and it is in terms of this relation that we shall interpret Eq. (8.75). If the reader wishes, he can view Eq. (8.75) as an equation in Hilbert space, where the symbol $(,)$ has its usual interpretation as an inner product. However, it is by no means necessary to do so; the results which we will obtain below are more general.

Since we have already seen that the finite-dimensional Theorem 4.27 holds for separable kernels, it will be interesting to see if it also holds in this more general case. We therefore want to consider the adjoint transformation, $K_F^\dagger$, which is defined by

$$K_F^\dagger \tilde{f} = \sum_{n=1}^{N} \psi_n (\phi_n, \tilde{f}) \ . \tag{8.76}$$

In the light of our previous discussion of Banach spaces, it is clear that if K_F acts on the space L_p, transforming $f \in L_p$ into $K_F f \in L_p$, then $K_F^\dagger$ acts on the space L_q $(1/p + 1/q = 1)$, transforming $\tilde{f} \in L_q$ into $K_F^\dagger \tilde{f} \in L_q$. We may remark at this point that seemingly more general forms for K_F such as, for example,

$$K_F' f \equiv \sum_{m,\, n=1}^{N} k_{mn} \phi_m (\psi_n, f) \ ,$$

where k_{mn} is a collection of constants, are included in our definition of Eq. (8.75) since ϕ_m and ψ_n are not assumed to be normalized in any particular way and N, the range of the sum, is completely arbitrary.

We now consider, in the spirit of Theorem 4.27, the two equations

$$f - \lambda K f = g \tag{8.77}$$

and

$$\tilde{h} - \lambda^* K^\dagger \tilde{h} = 0 \ , \tag{8.78}$$

where f and g are vectors in B and $\tilde{h}$ is a vector in the dual space of B. Using the definitions of Eqs. (8.75) and (8.76) respectively, we get

$$f = g + \lambda \sum_{n=1}^{N} \phi_n (\psi_n, f) \tag{8.79}$$

and

$$\tilde{h} - \lambda^* \sum_{n=1}^{N} \psi_n (\phi_n, \tilde{h}) = 0 \ . \tag{8.80}$$

Equation (8.79) tells us that we need to determine only the collection of constants,

$$a_n \equiv (\psi_n, f) \ ,$$

to obtain a solution. Using Eq. (8.79), we find that

$$(\psi_m, f) = (\psi_m, g) + \lambda \sum_{n=1}^{N} (\psi_m, \phi_n)(\psi_n, f) .$$

Setting

$$\alpha_{mn} \equiv (\psi_m, \phi_n) , \qquad b_m \equiv (\psi_m, g) , \tag{8.81}$$

this becomes

$$a_m - \lambda \sum_{n=1}^{N} \alpha_{mn} a_n = b_m .$$

It is convenient to write this in matrix notation as

$$(I - \lambda A)a = b , \tag{8.82}$$

where

$$[A]_{mn} = \alpha_{mn} , \qquad [a]_n = a_n , \qquad [b]_n = b_n .$$

Similarly, if we set $\tilde{a}_n = (\phi_n, \tilde{h})$, the homogeneous adjoint equation is

$$\tilde{a}_m - \lambda^* \sum_{n=1}^{N} (\phi_m, \psi_n)\tilde{a}_n = 0 ,$$

which, according to Eq. (8.81), may be written as

$$\tilde{a}_m - \lambda^* \sum_{n=1}^{N} \alpha_{nm}^* \tilde{a}_n = 0 .$$

In matrix notation, this is just

$$(I - \lambda A)^\dagger \tilde{a} = 0 , \tag{8.83}$$

where $[\tilde{a}]_n = \tilde{a}_n$. Thus our basic equations, Eqs. (8.77) and (8.78), have been transformed into the matrix equations, Eqs. (8.82) and (8.83).

But Eqs. (8.82) and (8.83) are in just the form required by Theorem 4.27. We conclude, therefore, that Eq. (8.82) has a solution if and only if

$$(b, \tilde{a}) = 0 = \sum_{n=1}^{N} b_n^* \tilde{a}_n \tag{8.84}$$

for all $\tilde{a}$ which are solutions of the homogeneous adjoint equation. Similarly, Eq. (8.77) will have solutions only under these conditions. Using the above definitions of b_n and $\tilde{a}_n$, this becomes

$$\sum_{n=1}^{N} (g, \psi_n)(\phi_n, \tilde{h}) = 0 .$$

But by our definition of $K^\dagger$ [Eq. (8.76)], this is just

$$(g, K^\dagger \tilde{h}) = 0 .$$

If $\tilde{h} \not\equiv 0$, $\lambda^* K^\dagger \tilde{h} = \tilde{h}$, by Eq. (8.78), so a solution to Eq. (8.77) exists if we require that

$$(g, \tilde{h}) = 0 \tag{8.85}$$

for all $\tilde{h}$ satisfying Eq. (8.78). Note that if Eq. (8.83) has no solution then $[I - \lambda A]^{-1}$ exists and Eq. (8.82) has a *unique* solution. Thus, if Eq. (8.78) has no solution, then Eq. (8.77) has a unique solution. We have, therefore, the following result:

Theorem 8.1. If K_F is a linear transformation of finite rank, then the equation

$$f = g + \lambda K_F f$$

has a solution if and only if $(g, \tilde{h}_i) = 0$ for all i, where $\tilde{h}_i$ satisfies the homogeneous adjoint equation

$$\lambda^* K_F^\dagger \tilde{h}_i = \tilde{h}_i .$$

If the homogeneous adjoint equation has no solutions then the solution to the inhomogeneous equation is unique.

By an identical argument, the same result holds for $K_F^\dagger$.

Corollary. If K_F is a linear operator of finite rank, then the equation

$$\tilde{f} = \tilde{g} + \lambda^* K_F^\dagger \tilde{f}$$

has a solution if and only if $(\tilde{g}, h_i) = 0$, where h_i is any solution of the homogeneous equation

$$\lambda K_F h_i = h_i .$$

If the homogeneous equation has no solution, then the solution to the inhomogeneous adjoint equation is unique.

The next corollary is an immediate consequence of the finite-dimensional Theorem 4.26.

Corollary. The equations

$$\lambda K_F h_i = h_i \qquad \text{and} \qquad \lambda^* K_F^\dagger \tilde{h}_i = \tilde{h}_i$$

have the same number of linearly independent solutions.

Theorem 8.1 and its two corollaries are often referred to as the Fredholm alternative ("alternative" because it expresses on option—*either* the inhomogeneous linear equation, $(I - \lambda K)f = g$, has a unique solution *or* the homogeneous adjoint equation possesses at least one solution). They represent an extension of the theorems of Section 4.8 to an infinite-dimensional space, but only for operators of a rather special type. However, in the next section we shall see that this result can be extended to a much broader class of operators.

Before extending these results, let us remark that if a unique solution to Eq. (8.77) exists, then we can find it explicitly by using Eqs. (8.79) and (8.82).

Equation (8.82) tells us that

$$a = (I - \lambda A)^{-1} b ,$$

or, in component form

$$\begin{aligned} a_n \equiv (\phi_n, f) &\equiv \sum_{m=1}^{N} [(I - \lambda A)^{-1}]_{nm} b_m \\ &= \sum_{m=1}^{N} [(I - \lambda A)^{-1}]_{nm} (\phi_m, g) . \end{aligned}$$

Putting this result into Eq. (8.79), we find that

$$f = g + \lambda \sum_{n=1}^{N} \sum_{m=1}^{N} \phi_n [(I - \lambda A)^{-1}]_{nm} (\phi_m, g) , \tag{8.86}$$

which gives us the desired solution if the equation

$$\lambda^* K_F^\dagger \bar{h} = \bar{h}$$

has no solutions, or, equivalently, if the equation

$$(I - \lambda A)^\dagger \bar{a} = 0$$

has no solutions. Thus the problem of solving Eq. (8.77) reduces to calculating the inverse of a given matrix of scalars. Equation (8.86) may be thought of as defining a new operator, G_{K_F}, which is the inverse of $(I - \lambda K_F)$. According to Eq. (8.86), we write

$$G_{K_F} \equiv I + \lambda R_{K_F} = (I - \lambda K_F)^{-1} , \tag{8.87}$$

where

$$R_{K_F} f \equiv \sum_{n=1}^{N} \sum_{m=1}^{N} [(I - \lambda A)^{-1}]_{mn} \phi_m (\phi_n, f) ; \tag{8.88}$$

R_{K_F} is called the *resolvent* operator corresponding to K_F. It is clearly also an operator of finite rank. It exists whenever $(I - \lambda A)^{-1}$ exists, that is, whenever

$$\det (I - \lambda A) \neq 0 .$$

Since $\det (I - \lambda A)$ is a polynomial of degree N in λ, and therefore has N (not necessarily distinct) roots, we see that R_{K_F} is a *meromorphic function* of λ—that is, it is analytic everywhere except for a finite number of poles. At these poles, R_{K_F} fails to exist, and we have a solution to the equation

$$\lambda K_F h = h ,$$

or, equivalently, to the equation

$$\lambda^* K_F^\dagger \bar{h} = \bar{h} .$$

It is a worthwhile exercise to check that

$$(I - \lambda K_F)(I + \lambda R_{K_F}) = (I + \lambda R_{K_F})(I - \lambda K_F) = I .$$

As an example of a kernel of finite rank, consider the equation

$$f(x) = x + \lambda \int_0^\pi \frac{\sin(x + x')}{\pi} f(x')\, dx'$$

$$= x + \lambda \frac{\sin x}{\sqrt{\pi}} \int_0^\pi \frac{\cos x'}{\sqrt{\pi}} f(x')\, dx' + \lambda \frac{\cos x}{\sqrt{\pi}} \int_0^\pi \frac{\sin x'}{\sqrt{\pi}} f(x')\, dx' . \quad (8.89)$$

Calling

$$a_1 = \frac{1}{\sqrt{\pi}} \int_0^\pi \cos x' f(x')\, dx' , \qquad a_2 = \frac{1}{\sqrt{\pi}} \int_0^\pi \sin x' f(x')\, dx' ,$$

and

$$b_1 = \frac{1}{\sqrt{\pi}} \int_0^\pi x \cos x\, dx , \qquad b_2 = \frac{1}{\sqrt{\pi}} \int_0^\pi x \sin x\, dx ,$$

We find that Eq. (8.89) may be transformed to

$$a_1 = b_1 + \lambda \frac{1}{\pi} \int_0^\pi \cos^2 x\, dx\, a_2 , \qquad a_2 = b_2 + \lambda \frac{1}{\pi} \int_0^\pi \sin^2 x\, dx\, a_1 .$$

Thus in matrix notation

$$a = b + \lambda M a \qquad \text{or} \qquad (I - \lambda M) a = b ,$$

where

$$M = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}, \qquad a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \frac{1}{\sqrt{\pi}} \begin{bmatrix} -2 \\ \pi \end{bmatrix} .$$

The inverse matrix $(I - \lambda M)^{-1}$ is

$$\begin{bmatrix} 1 & -\lambda/2 \\ -\lambda/2 & 1 \end{bmatrix}^{-1} = \frac{1}{1 - \lambda^2/4} \begin{bmatrix} 1 & \lambda/2 \\ \lambda/2 & 1 \end{bmatrix} .$$

Hence

$$a = (I - \lambda M)^{-1} b = \frac{1}{1 - \lambda^2/4} \begin{bmatrix} 1 & \lambda/2 \\ \lambda/2 & 1 \end{bmatrix} \frac{1}{\sqrt{\pi}} \begin{bmatrix} -2 \\ \pi \end{bmatrix} ,$$

so

$$a_1 = \frac{2}{\sqrt{\pi}} \frac{(\pi\lambda - 4)}{4 - \lambda^2} , \qquad a_2 = \frac{4}{\sqrt{\pi}} \frac{\pi - \lambda}{4 - \lambda^2} .$$

Thus

$$f(x) = x + 2\lambda \frac{(\lambda - 4/\pi)}{4 - \lambda^2} \sin x + 4\lambda \frac{(1 - \lambda/\pi)}{4 - \lambda^2} \cos x . \quad (8.90)$$

Clearly, when $\lambda = \pm 2$, this solution becomes infinite, corresponding to the existence of solutions to the homogeneous equation for these values.

The Neumann series for $f(x)$ is

$$f(x) = x + \frac{\lambda}{\pi}\int_0^\pi dx' \sin(x + x')x' + \frac{\lambda^2}{\pi^2}\int_0^\pi dx' \int_0^\pi dx'' \sin(x + x')\sin(x' + x'')x'' + \cdots .$$

According to our previously developed convergence criteria, this series will certainly converge when

$$\frac{|\lambda|}{\pi}\left[\int_0^\pi \int_0^\pi \sin^2(x + x')\, dx\, dx'\right]^{1/2} < 1 .$$

Performing the integrations, we find the requirement that $\lambda < \sqrt{2}$. The first few terms of the Neumann series, written out explicitly, are

$$f(x) = x + \lambda[\cos x - (2/\pi)\sin x] + \lambda^2[(1/2)\sin x - (1/\pi)\cos x] + \cdots \tag{8.91}$$

This result agrees with the first few terms in the power series expansion of Eq. (8.90). Since the power series converges whenever $\lambda < 2$, it is clear that the Neumann series must also converge if $\lambda < 2$. Thus we see that the estimate on the norm of an integral operator given by the integrated square of its kernel provides only a lower bound on the radius of convergence of the Neumann series. In fact, even if we knew the exact norm of the operator, the requirement $||K|| < 1$ only tells us that the Neumann series *will* converge if $||K|| < 1$, not that it will *fail* to converge if $||K|| > 1$.

When $\lambda = \pm 2$, the two eigenvectors of

$$f(x) = \lambda \int_0^\pi \frac{\sin(x + x')}{\pi} f(x')\, dx'$$

turn out to be $f_+(x) = A_+ (\sin x + \cos x)$ and $f_-(x) = A_- (\sin x - \cos x)$, corresponding respectively to $\lambda_+ = 2$ and $\lambda_- = -2$. If we normalize them to unity, we find that

$$f_+(x) = \sqrt{\frac{2}{\pi}} \sin\left(x + \frac{\pi}{4}\right), \qquad f_-(x) = \sqrt{\frac{2}{\pi}} \sin\left(x - \frac{\pi}{4}\right).$$

Note that these two solutions are orthogonal, as they should be, since $k(x, y) = (1/\pi)\sin(x + y)$ is a real symmetric kernel.

Figure 8.1 shows, for the case $\lambda = 1$, the exact solution compared to the iterates:

$$f_0(x) = x ,$$
$$f_1(x) = x + \lambda[\cos x - (2/\pi)\sin x] ,$$
$$f_2(x) = x + \lambda[\cos x - (2/\pi)\sin x] + \lambda^2[(1/2)\sin x - (1/\pi)\cos x] .$$

It is seen that $f_2(x)$ is a very good approximation to the exact solution, even though $||K_F|| \cong \frac{1}{2}$ when $\lambda = 1$.

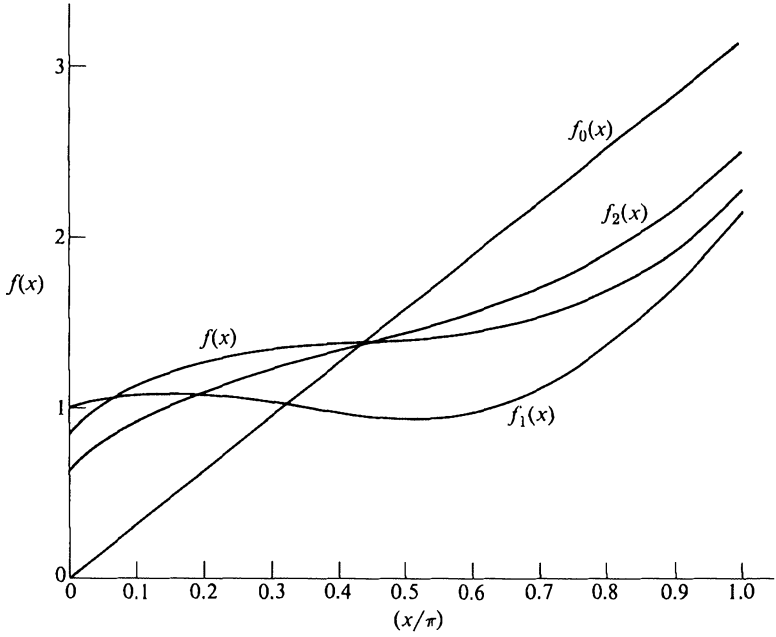

Fig. 8.1 The exact solution, $f(x)$, to Eq. (8.89) of the text, compared with the first three Neumann iterates, $f_0(x)$, $f_1(x)$, and $f_2(x)$.

8.7 COMPLETELY CONTINUOUS OPERATORS

Although the kernels of finite rank which we have discussed thus far lead to integral equations which can be solved in closed form, they are seldom *directly* relevant to physical problems. The integral equations which arise in physics (for example, in Schrödinger's integral equation and its relativistic generalizations) usually have kernels, $k(x, y)$, in which the x- and y-dependence cannot be unravelled into sums of the form of Eq. (8.75).

The reason that we have nevertheless spent so much time talking about these transformations of finite rank is explained by the following theorem.

Theorem 8.2. Any linear transformation of Hilbert space whose kernel is continuous and square integrable as a function of two variables can be approximated arbitrarily closely in norm by a linear transformation of finite rank.

Proof. The simplest case is when the functions in the Hilbert space are defined on a finite interval. Then the two-dimensional Weierstrass theorem tells us that $k(x, y)$ can be approximated *uniformly* by a polynomial; in other words, given any δ there exists a polynomial $P_N(x, y)$, where N depends on δ such that

$$|k(x, y) - P_N(x, y)| < \delta$$

for all x, y. Note that a kernel in polynomial form is a kernel of finite rank since

$$P_N(x, y) = \sum_{m, n=0}^{N} C_{mn} x^m y^n$$

for all x, y. Hence

$$\int_a^b dx \int_a^b dy \, |k(x, y) - P_N(x, y)| \leq [\delta(b - a)]^2 .$$

Thus we can find an N which will provide a δ to make

$$||K - K_N|| < \epsilon$$

for any preassigned ϵ, where K_N is the transformation whose kernel is $P_N(x, y)$. Now if the Hilbert space is defined over an infinite interval, we need only truncate the kernel by setting it equal to zero for all x, y which lie outside some set, B, of finite area. Call this kernel k_B and the related linear transformation K_B. Then if we let $C(B)$ denote the complement of B in the xy-plane,

$$\int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \, |k(x, y) - k_B(x, y)|^2 = \int_{C(B)} dx \int dy \, |k(x, y)|^2 .$$

But $k(x, y)$ is square integrable, so it must be possible to choose a set of points outside of which the integral of $|k|^2$ is arbitrarily small. Thus we can choose B so that

$$||K - K_B|| < \epsilon/2 .$$

But by previous reasoning, we can find a polynomial kernel $k_N(x, y)$ with related linear transformation K_N such that

$$||K_B - K_N|| < \epsilon/2 .$$

Thus

$$||K - K_N|| \leq ||K - K_B|| + ||K_B - K_N|| < \epsilon$$

for a suitable choice of N, which is what we wanted to show.

This theorem can easily be weakened to include the case of piecewise continuous kernels. Furthermore, the theorem can be extended to the class of square-integrable kernels with no continuity or boundedness requirements on $k(x, y)$ other than those implied by square integrability. The only difficulty in this more general case will be the fact that the kernel can become infinite at some points. For example, in the space $L_2[0, 1]$, the linear operator whose kernel is

$$k(x, y) = 1/\sqrt{x + y} \tag{8.92}$$

is an admissible operator because $k(x, y)$ is square integrable. However, at the point $x = 0$, $y = 0$ the kernel becomes infinite. This difficulty is easily overcome. We truncate the kernel by setting it equal to zero whenever $|k(x, y)| \geq$

L; where L is a number which may be made as large as we please. The truncated kernel, $k_L(x, y)$, can be approximated in the manner of Theorem 8.2. However, since $k(x, y)$ is square integrable, it must be possible to choose L sufficiently large so that we make negligible error in using $k_L(x, y)$ instead of $k(x, y)$; that is, the number

$$I_L = \int dx \int dy \, |k(x, y) - k_L(x, y)|^2$$

can be made as small as we please. I_L can be written as

$$I_L = \iint_{\mathscr{D}_L} |k(x, y)|^2 \, dx \, dy \, ,$$

where $\mathscr{D}_L$ is the set of points on which $|k(x, y)| \geq L$. Now clearly it must be possible to choose L sufficiently large so that the area of $\mathscr{D}_L$ is as small as we please [otherwise $k(x, y)$ would not be square integrable]. Also, it is clear that I_L must be finite for any $\mathscr{D}_L$ or again $k(x, y)$ would not be square integrable. There remains only the possibility that although the area of $\mathscr{D}_L$ is as small as we please, the integral of $|k(x, y)|^2$ over this arbitrarily small set is finite. Now it is possible to imagine objects which are singular in just such a way that their integral over an arbitrarily small set is finite; the δ-function, for example, has this property. However, in the Lebesgue theory of integration (and also in the Riemann theory) such objects are excluded from the class of admissible *functions*, and therefore we exclude them from the vector space of square-integrable functions. The kernel mentioned above [Eq. (8.92)], although singular at the origin, still has the property that the integral of $|k(x, y)|^2$ over a region of arbitrarily small area containing the origin tends toward zero as the area of the region tends toward zero. Thus *Theorem 8.2 can be extended to the space of square-integrable functions.*

Because of the practical importance of square-integrable kernels (see, for example, Section 8.3) this result is extremely useful, and in Chapter 9 we will develop its consequences in considerable detail. However, for many purposes it is best to concentrate not on the property of finite rank approximation, but rather on a more subtle property which can be inferred from it (and which is, in fact, equivalent to it). It is with this in mind that we introduce the concept of a *completely continuous* operator.

Definition 8.8. A linear operator on a normed vector space is said to be *completely continuous* if it transforms *any* bounded sequence of vectors (i.e., vectors whose norms are all less than some fixed constant) into a sequence containing a subsequence which converges in norm.

The use of the terminology *completely continuous* is fairly clear from the definition. An *ordinary* continuous (i.e., bounded) transformation maintains the status quo regarding convergent sequences; if $\{f_n\}$ converges, so does $\{Kf_n\}$ if K is continuous. But a *completely* continuous transform *forces* sequences to converge, even though they may not have converged originally. For the defini-

tion of complete continuity to be at all interesting, it must be possible to choose an infinite sequence of vectors which has a subsequence that does not converge. In a finite-dimensional space, it is impossible to find such a sequence because of the existence of the Bolzano-Weierstrass theorem* for real numbers. Thus the definition is empty on a finite-dimensional space; or if one likes, one may say that on a finite-dimensional vector space all operators are completely continuous.

Let us obtain some of the simpler properties of such transformations. Obviously, the sum of any finite number of completely continuous transformations is completely continuous. We can also show that the product of a completely continuous transformation and a bounded transformation is completely continuous. Suppose that K is completely continuous and let B be a bounded transformation. Consider KB acting on $\{f_n\}$, where $||f_n|| < C$ for all n. Then K acts on the sequence $\{Bf_n\}$, which is bounded because B is a bounded operator. Since K is completely continuous, there must be a subsequence of $\{KBf_n\}$ which converges in norm to an element of the space, so KB is completely continuous. Now consider BK acting on $\{f_n\}$. Since K is completely continuous, $\{Kf_n\}$ possesses a subsequence which converges in norm. Since B is bounded, it is continuous, so $\{BKf_n\}$ has a trivially related convergent subsequence.

We now show that the class of such transformations is not empty. Consider the space of square-integrable functions and take the separable transformation K_s with kernel

$$k_s(x, y) = \phi(x)\psi^*(y) ,$$

along with any sequence $\{f_n\}$, $||f_n|| < C$ for all n. Now

$$K_s f_n = \phi(x)(\psi, f_n) .$$

Since $|(\psi, f_n)| \leq ||\psi|| \cdot ||f_n||$, the sequence of complex numbers (ψ, f_n) is bounded in the complex plane and therefore possesses a convergent subsequence (Bolzano-Weierstrass theorem). Call the limit of this subsequence l, and consider

$$||K_s f_{n_k} - l\phi|| = ||\phi[(\psi, f_{n_k}) - l]|| \leq ||\phi|| \cdot |(\psi, f_{n_k}) - l| ,$$

where n_k $(k = 1, 2, 3, \cdots)$ defines the convergent subsequence. Since $|(\psi, f_{n_k}) - l|$ tends toward zero as n_k increases, we have

$$K_s f_{n_k} \to l\phi$$

in the norm of the space, so K_s is completely continuous. Hence any operator of finite rank is completely continuous, since it is a finite sum of separable operators.

Of course, there are also many operators which are not completely continuous. Consider $D_x = -id/dx$ acting on the space of square-integrable functions on [0, 1]. We know that $\{f_n = e^{inx}\}$ is a bounded sequence, but $\{Df_n =$

* For our purposes, the Bolzano-Weierstrass theorem may be stated as follows: From every bounded sequence of points in n-dimensional Euclidean space, it is possible to select a convergent subsequence.

$ne^{inx}\}$ does not contain a convergent subsequence. In fact, even the identity operator is not completely continuous.

A particularly interesting property of the class of completely continuous operators is that the limit of any sequence of completely continuous operators is itself a completely continuous operator.

Theorem 8.3. If $||K_i - K|| \to 0$ as $i \to \infty$, and if K_i is a completely continuous transformation for all i, then K is also completely continuous.

Proof. Let $\{f_n\}$ be any bounded sequence of functions. Then we can select a subsequence, $\{f_{n_1}^{(1)}\}$, such that $\{K_1 f_{n_1}^{(1)}\}$ converges in norm, since K_1 is completely continuous. Now consider the bounded sequence $\{f_{n_1}^{(1)}\}$. From this we can select a subsequence, $\{f_{n_2}^{(2)}\}$, such that $\{K_2 f_{n_2}^{(2)}\}$ converges in norm. Clearly, $\{K_1 f_{n_2}^{(2)}\}$ will still converge in norm. We can continue this process indefinitely, thus obtaining finally a sequence $\{\phi_m\}$ such that $\{K_i\phi_m\}$ converges in norm for all K_i. Now consider

$$K\phi_n - K\phi_m = K\phi_n - K_i\phi_n + K_i\phi_n - K_i\phi_m + K_i\phi_m - K\phi_m .$$

By the triangle inequality, we find that

$$\begin{aligned} ||K\phi_n - K\phi_m|| &\leq ||K\phi_n - K_i\phi_n|| + ||K_i\phi_n - K_i\phi_m|| + ||K_i\phi_m - K\phi_m|| \\ &\leq ||K - K_i||\cdot||\phi_n|| + ||K_i\phi_n - K_i\phi_m|| + ||K - K_i||\cdot||\phi_m|| . \end{aligned}$$

Since the ϕ_n are bounded,

$$||K\phi_n - K\phi_m|| \leq C||K - K_i|| + ||K_i\phi_n - K_i\phi_m|| .$$

Because $||K - K_i|| \to 0$, we can pick some i sufficiently large so that

$$||K - K_i|| < \epsilon/(2C) .$$

Then when $m, n > N$ the second term on the right above can be made smaller than $\epsilon/2$. Thus for $m, n > N$, we have

$$||K\phi_n - K\phi_m|| < \epsilon .$$

Hence the sequence $\{K\phi_n\}$ is convergent, and the proof is complete.

As we remarked above, any transformation with a square-integrable kernel can be approximated arbitrarily closely in norm by a kernel of finite rank. Hence such transformations are completely continuous. This theorem underlines the fact that the projection operators, P_n, defined in Section 8.2 do not converge in norm to I, the identity operator. If they did, then since P_n is of finite rank and hence completely continuous, Theorem 8.3 would say that the identity operator is completely continuous, which it is not.

The above results show that any operator which can be approximated in norm by a sequence of operators of finite rank is completely continuous. We now want to show that the converse is also true, namely, any completely continuous operator can be approximated in norm arbitrarily closely by a sequence of operators of finite rank. In order that we may make use of a complete orthonormal basis, we will prove this result only in Hilbert space.

Patterning our work on the techniques of Chapter 5, let us imagine that we are dealing with an integral transformation whose kernel is $k(x, y)$. Let $\{\phi_n\}$ be an orthonormal basis in L_2. We write

$$k(x, y) = \sum_{n, m=1}^{\infty} (\phi_n \phi_m^*, k)\, \phi_n(x)\, \phi_m^*(y) ,$$

in the sense of mean convergence. Here we make use of the fact that the collection of all products $\phi_n(x)\phi_m(y)$ [or $\phi_n(x)\phi_m^*(y)$] forms a complete orthonormal basis for the Hilbert space of square-integrable functions of two variables to which $k(x, y)$ belongs. By truncating the sum at some large N, we obtain a kernel of finite rank

$$k_N(x, y) = \sum_{n, m=1}^{N} (\phi_n \phi_m^*, k)\, \phi_n(x)\, \phi_m^*(y) .$$

One would hope, on the basis of experience with Fourier series, that this will approximate $k(x, y)$ arbitrarily closely in norm for N sufficiently large. The related transformation K_N is defined by

$$K_N f = \sum_{n, m=1}^{N} \phi_n(x)\, (\phi_n \phi_m^*, k)\, (\phi_m, f) . \tag{8.93}$$

We can abbreviate this by writing K_N as (see Problem 8.26)

$$K_N = P_N K P_N ,$$

where P_N is the projection operator defined by

$$P_N f \equiv \sum_{n=1}^{N} (\phi_n, f)\phi_n . \tag{8.94}$$

With the above discussion as background, we now state the final theorem of this chapter in completely generality.

Theorem 8.4. Let $\{\phi_n\}$ be an orthonormal basis* for a Hilbert space H, and let K be a completely continuous operator on H. If we denote by P_N the operator that projects onto the subspace of H spanned by $\{\phi_1, \phi_2, \cdots, \phi_N\}$ and define $K_N \equiv P_N K P_N$, then the operator K_N tends in norm to K as N tends to infinity.

Proof. Consider

$$\begin{aligned} K - K_N &= K - P_N K P_N = K - P_N K + P_N K - P_N K P_N \\ &= (I - P_N)K + P_N K(I - P_N) . \end{aligned} \tag{8.95}$$

---

* We are assuming here the existence of a *countable* basis for H. A Hilbert space with this property is called a *separable* Hilbert space; most, but not all, spaces which occur in physics are of this type. The proof of Theorem 8.4 can be extended to nonseparable Hilbert spaces.

If it were true that P_N tended toward I in operator norm, then the triangle inequality applied to Eq. (8.95) would complete the proof, which would then hold for *any bounded* operator. However, as we have already seen, this is *not* the case. The result will be proved by contradiction. The basic idea is that if the theorem is *not* true, then there must exist some real number a, such that for each n we can find a vector f_n, which satisfies

$$||(K - K_n)f_n|| > a||f_n|| , \tag{8.96}$$

where a is independent of n and $a > 0$. If there were no such number a, then according to Definition 8.7, K_n would tend toward K in norm. If we divide both sides of Eq. (8.96) by $||f_n||$, we may say that if the theorem is *not* true then there exists a sequence of functions $\{f_n: ||f_n|| = 1\}$ such that

$$||(K - K_n)f_n|| > a \tag{8.97}$$

for all n. Assuming that K is completely continuous, we will show that Eq. (8.97) does not hold, so the theorem must be true. It is important to appreciate the fact that it is *not* enough to show that for any fixed $f \in H$, $||(K - K_n)f|| \to 0$. The triangle inequality applied to Eq. (8.95) would prove this.

We begin our proof by using the triangle inequality together with the relation $||P_N|| \leq 1$, to obtain from Eq. (8.95)

$$\begin{aligned} ||(K - K_N)f_N|| &\leq ||(I - P_N)Kf_N|| + ||P_N K(I - P_N)f_N|| \\ &\leq ||(I - P_N)Kf_N|| + ||K(I - P_N)f_N|| . \end{aligned} \tag{8.98}$$

We must now show that both $||(I - P_N)Kf_N||$ and $||K(I - P_N)f_N||$ become small as N increases. It is at this point that the complete continuity of K becomes of central importance. It enables us to choose from the sequence $\{f_N\}$ a subsequence $\{\bar{g}_n\}$ such that $\{K\bar{g}_n\}$ converges to a limit, say $\xi \in H$. Consider also the sequence $\{(I - P_n)\bar{g}_n\}$, which is a bounded sequence since $||I - P_n|| = 1$ and $||\bar{g}_n|| = 1$. Because K is completely continuous, we can choose a subsequence $\{(I - P_m)g_m\}$, such that $\{K(I - P_m)g_m\}$ converges to a limit, say $\eta \in H$. Note that $\{Kg_m\}$ will converge to ξ since $\{K\bar{g}_n\}$ converges to ξ and $\{g_m\}$ is a subsequence of $\{\bar{g}_n\}$. In order to avoid confusion over notation, let us remind the reader that each g_m is equal to some element of our original collection $\{f_N\}$.

The meat of the proof is now behind us; only a few estimates of $||(I - P_m)Kg_m||$ and $||K(I - P_m)g_m||$ remain to be made. Consider first $||(I - P_m)Kg_m||$. We write

$$(I - P_m)Kg_m \equiv (I - P_m)(Kg_m - \xi) + (I - P_m)\xi ,$$

where ξ is a fixed vector, and using the triangle inequality, we find that

$$\begin{aligned} ||(I - P_m)Kg_m|| &\leq ||(I - P_m)(Kg_m - \xi)|| + ||(I - P_m)\xi|| \\ &\leq ||Kg_m - \xi|| + ||\xi - P_m\xi|| , \end{aligned}$$

since $||I - P_m|| = 1$. But by construction, Kg_m tends in norm toward ξ, and since ξ is fixed, $P_m\xi$ tends toward ξ in norm. Thus

$$||(I - P_m)Kg_m|| \to 0 \tag{8.99}$$

as $m \to \infty$.

Finally, we discuss $||K(I-P_m)g_m||$. By construction, as $m \to \infty$, $K(I-P_m)g_m$ tends in norm toward η. We will show that $||\eta|| = 0$. Consider

$$||\eta||^2 = (\eta, \eta) = \lim_{m\to\infty} (\eta, K(I-P_m)g_m) = \lim_{m\to\infty} ((I-P_m)K^\dagger\eta, g_m) ,$$

where we have used that fact that P_m is self-adjoint (regarding the second equality in the above equation; see Problem 8.21a). But

$$|((I-P_m)K^\dagger\eta, g_m)| \leq ||(I-P_m)K^\dagger\eta|| \, ||g_m|| ,$$

by Schwartz's inequality, so using $||g_m|| = 1$, we have

$$||\eta||^2 \leq \lim_{m\to\infty} ||(I-P_m)K^\dagger\eta|| .$$

Since $K^\dagger\eta$ is a fixed vector, $||(I-P_m)K^\dagger\eta|| \to 0$ as $m \to \infty$, so $||\eta|| = 0$. Thus η is the zero vector, and since $K(I-P_m)g_m$ tends in norm toward η, we may write

$$||K(I-P_m)g_m|| \to 0 \tag{8.100}$$

as $m \to \infty$. Putting the results of Eqs. (8.99) and (8.100) into Eq. (8.98), we see that

$$||(K-K_m)g_m|| \to 0$$

as $m \to \infty$. Since the g_m form a subset of our original collection of f_N, *all* of which were assumed to satisfy $||(K-K_N)f_N|| > a$, where a is strictly positive, we see that we have reached the desired contradiction. Thus the theorem is proved.

This theorem suggests a very attractive method for solving equations involving completely continuous transformations. We just split any such transformation into a "small" operator plus an operator of finite rank. The former will yield to simple iterative techniques while the latter can be handled as an ordinary matrix, as we have seen before. In practice, many problems in physics and engineering must be handled in this way, since the operators are neither "small" enough for a simple Born approximation to be valid, nor simple enough to fall into the finite-rank category. The main problem that remains is to develop a method whereby the two processes, iteration and finite-rank approximation, can be combined. It is to this matter that Chapter 9 is primarily devoted.

PROBLEMS

1. Prove that the space C of continuous functions on $[a, b]$ with norm given by

$$||f|| = \max_{a\leq x\leq b} |f(x)|$$

for $f \in C$ is a Banach space, i. e., show that it is complete with respect to the above norm.

2. *Hölder's Inequality*

a) Show that the function $f(z) = z^\alpha - \alpha z - \beta$ takes on a maximum value at $z = 1$ if $\alpha < 1$ and that if $\beta = 1 - \alpha$, then this maximum value is zero. Thus $z^\alpha \leq \alpha z + \beta$ with $\alpha < 1$, $\beta = 1 - \alpha$.

b) By making use of the change of variables $z = x/y$, show that $x^\alpha y^\beta \leq \alpha x + \beta y$, where again $\alpha \leq 1$ and $\beta = 1 - \alpha$.

c) Suppose now that $|f(x)|^p$ and $|g(x)|^q$ are integrable functions with $1/p + 1/q = 1$. Use the result of part (b) to show that

$$|fg| \leq \frac{1}{p}|f|^p + \frac{1}{q}|g|^q,$$

so that $|fg|$ is also integrable.

d) Show that if $|f(x)|^p$ and $|g(x)|^q$ are integrable, then

$$\left|\int f(x)g(x)\,dx\right| \leq \left[\int |f(x)|^p\,dx\right]^{1/p}\left[\int |g(x)|^q\,dx\right]^{1/q}.$$

Hint: Apply part (*c*) to the functions

$$F(x) = f(x)/\left[\int |f(x)|^p\,dx\right]^{1/p} \quad \text{and} \quad G(x) = g(x)/\left[\int |g(x)|^q\,dx\right]^{1/q}.$$

This result generalizes Schwartz's inequality of Hilbert space and is known as Hölder's inequality.

3. *Minkowski's Inequality*

a) Show that $|f(x) + g(x)|^p \leq 2^p[|f(x)|^p + |g(x)|^p]$, and hence that if f and g belong to L_p (the space of p-integrable functions), then so does $f + g$. Thus L_p is indeed a vector space.

b) Show that for $p > 1$,

$$\int |f(x) + g(x)|^p\,dx \leq \int |f(x)|\,|f(x) + g(x)|^{p-1}\,dx + \int |g(x)|\,|f(x) + g(x)|^{p-1}\,dx,$$

and that $|f + g|^{p-1}$ belongs to L_q if f and g belong to L_p, where $1/p + 1/q = 1$.

c) Apply Hölder's inequality (see Problem 8.2) to each of the two integrals on the right-hand side of the inequality in part (*b*), and hence show that

$$\left[\int |f(x) + g(x)|^p\,dx\right]^{1/p} \leq \left[\int |f(x)|^p\,dx\right]^{1/p} + \left[\int |g(x)|^p\,dx\right]^{1/p}.$$

This is Minkowski's inequality. We see therefore that $[\int |f(x)|^p\,dx]^{1/p}$ does indeed define a norm for the space L_p; Minkowski's inequality is just a statement of the triangle inequality which is the essential ingredient in the definition of a norm.

4. Solve the equations

a) $f(x) = 1 + \lambda \int_0^\infty e^{-(x+y)} f(y)\,dy$ b) $f(x) = 1 + \lambda \int_0^x f(y)\,dy$

by iteration. For what values of λ will the iterations converge?

5. Multiplying both sides of Eq. (8.27) by $V^{1/2}(\mathbf{r})$ [assume $V(\mathbf{r})$ to be strictly positive], show that if we define

$$\tilde{\psi}(\mathbf{r}) = V^{1/2}(\mathbf{r})\psi(\mathbf{r}), \qquad \tilde{\xi}(\mathbf{r}) = V^{1/2}(\mathbf{r})\xi(\mathbf{r})$$

and

$$\tilde{k}(\mathbf{r}, \mathbf{r}', E) = V^{1/2}(\mathbf{r})G_0(\mathbf{r}, \mathbf{r}', E)V^{1/2}(\mathbf{r}') ,$$

then

$$\tilde{\psi}(\mathbf{r}) = \tilde{\xi}(\mathbf{r}) + \int \tilde{k}(\mathbf{r}, \mathbf{r}', E)\tilde{\psi}(\mathbf{r}')\, d\mathbf{r}' .$$

Calling $\tilde{K}$ the integral operator whose kernel is $\tilde{k}(\mathbf{r}, \mathbf{r}', E)$, show that the norm of $\tilde{K}$ is the same as the norm of the kernel of Eq. (8.27) obtained by the weight-function method. Show also that if we iterate the equation for $\tilde{\psi}(\mathbf{r})$, the object $V^{1/2}(\mathbf{r})$ never actually appears in the calculation—the ordinary Born series for $\psi(\mathbf{r})$ is duplicated, apart from a multiplicative factor of $V^{1/2}(\mathbf{r})$.

6. Estimate the norm of the kernel of the Lippmann-Schwinger equation when $V(\mathbf{r}) = V_0 e^{-\mu r}$. For what values of V_0 will the Born series converge according to this estimate.

7. a) Suppose that we have an attractive δ-function potential in one dimension, $V(x) = -V_0\delta(x)$. Using the result of Problem 7.10, show that this potential will support one and only one bound state, no matter how large V_0 may be. Determine the binding energy and wave function of this state.

   b) Suppose that particles of positive energy E are incident on a δ-function potential, $V(x) = V_0\delta(x)$, in one dimension. What fraction of the particles is transmitted through the potential barrier and what fraction is reflected?

8. a) Using the result of Problem 7.10, show that if we have a potential $V(x)$ in one dimension which vanishes outside the interval $[-a/2, a/2]$, then, if we denote the wave function in the region $[-\infty, -a/2]$ by $\psi_1(x)$, the wave function in the region $[-a/2, a/2]$ by $\psi_2(x)$ and the wave function in $[a/2, \infty]$ by $\psi_3(x)$, then, calling $E = \hbar^2k^2/2m$ where $E > 0$ is the energy of the particle and m is its mass,

$$\psi_1(x) = e^{ikx} - \frac{im}{\hbar^2 k} e^{-ikx} \int_{-a/2}^{a/2} e^{ikx'} V(x')\psi_2(x')\, dx' ,$$

$$\psi_3(x) = e^{ikx}\left[1 - \frac{im}{\hbar^2 k}\int_{-a/2}^{a/2} e^{-ikx'} V(x')\, \psi_2(x')\, dx'\right],$$

and

$$\psi_2(x) = e^{ikx} - \frac{im}{\hbar^2 k}\int_{-a/2}^{a/2} e^{ik|x-x'|} V(x')\psi_2(x')\, dx' .$$

Thus we see that in order to determine the wave function in the region where the potential doesn't act, we need only know the wave function in the region where the potential does act.

   b) In part (a) above, assume $V(x) = V_0 =$ constant in the region of interest. Then

$$\psi_2(x) = e^{ikx} - \frac{imV_0}{\hbar^2 k}\int_{-a/2}^{a/2} e^{ik|x-x'|}\psi_2(x')\, dx' .$$

Show that the nth term in the iteration of this equation is less than $(mV_0a/\hbar^2k)^{n-1}$ in magnitude; so the iteration will converge if $(mV_0a/\hbar^2k) < 1$. Since $\hbar k =$

$p = mv$, this is equivalent to $V_0 a/hv < 1$. This result also holds for three-dimensional problems, although it is much more difficult to prove in that case. Note that the Hilbert space techniques used in the text give estimates for the convergence of the Born series in the three-dimensional case which are independent of the energy of the particle, whereas the result $V_0 a/hv < 1$ says that for *any* potential strength the Born series will converge if the energy of the particle is *sufficiently large*.

9. Using the weight-function method, show that in one dimension the iteration of the Lippmann-Schwinger equation will converge if

$$\frac{1}{hv}\int_{-\infty}^{\infty} V(x)\,dx < 1\,,$$

where $V(x)$ is the potential. Note that if the range of $V(x)$ is a, then

$$\int_{-\infty}^{\infty} V(x)\,dx = \bar{V}a\,,$$

where $\bar{V}$ is a mean potential strength. Thus this result is a simple generalization of the result of Problem 8(b). In that problem, the basic physical ideas are clearer, since there we can sharply delineate a region in which the potential acts and a region in which it does not act. The weight function becomes necessary because of the fact that most practical potentials have a nonvanishing magnitude at *all* points of space, although at *most* points the magnitude is small.

10. In Problem 8(a), assume that $V(x) = V_0 = \text{const}$, with V_0 small in the sense of Problem 8(b). Calculate ψ_1 and ψ_3 using the lowest-order approximation to ψ_2. What is the transmission probability in this approximation? What is the probability of reflection? Note that these two probabilities do not add up to 1. Why not? (For exact answers, see L. I. Schiff, *Quantum Mechanics*, McGraw-Hill, second edition, 1955, Section 17.)

11. Show that if $V(\mathbf{r})$ changes sign as $\mathbf{r}$ varies, then the appropriate modification of $\rho(\mathbf{r})$ in the weight-function method is to write $d\rho(\mathbf{r}) = |V(\mathbf{r})|d\mathbf{r}$. Obtain an estimate on the norm of the Lippmann-Schwinger integral operator using this weight function.

12. a) Making use of the definition $\psi(x) = e^{ikx}\phi(x)$ in the one-dimensional Lippmann-Schwinger equation, show that $\phi(x)$ satisfies the integral equation

$$\phi(x) = 1 - \frac{im}{h^2k}\int_{-\infty}^{x} V(x')\phi(x')\,dx' - \frac{im}{h^2k}e^{-ikx}\int_{x}^{\infty} e^{2ikx'}V(x')\phi(x')\,dx'\,.$$

b) Make some plausible arguments to suggest that the last term on the right-hand side of the above equation is negligible if k is large (i.e., if the energy of the incident particle is large).

c) Neglecting this term, show that the resulting $\phi(x)$, which we will call $\phi_E(x)$ satisfies the differential equation

$$\frac{d\phi_E(x)}{dx} = -\frac{im}{h^2k}V(x)\phi_E(x)\,,$$

with the boundary condition $\phi_E(-\infty) = 1$. Hence show that

$$\phi_E(x) = \exp\left\{-\frac{im}{h^2k}\int_{-\infty}^{x} V(x')\,dx'\right\}.$$

This approximation is referred to as the *eikonal* approximation; its generalization to three dimensions has proved very useful in the analysis of certain high-energy scattering problems.

d) Returning to parts (a) and (b), show that for the case $V(x) = V_0$ for $x \in [-a/2, a/2]$, $V(x) = 0$ otherwise, the last term of the equation in part (a) really is small as $k \to \infty$. Do this by replacing $\phi(x)$ in the integral by the approximate $\phi_E(x)$ given by part (c). The resulting integral is simple; show that this term is less in magnitude than

$$\frac{1}{2}\left|\frac{E}{V_0} - \frac{1}{4}\right|^{-1},$$

so that for $E \gg V_0$ the term is small.

13. Consider the differential equation

$$\frac{d^2x(t)}{dt^2} + 2\gamma\frac{dx(t)}{dt} + \omega_0^2 x(t) = 0$$

for a damped harmonic oscillator.

a) Show that if $x(0) = x_0$ and $\dot{x}(0) = 0$, then $x(t)$ satisfies the integral equation

$$x(t) = x_0 \cos \omega_0 t + \frac{2x_0\gamma}{\omega_0} \sin \omega_0 t - 2\gamma \int_0^t \cos \omega_0(t - t')x(t')\, dt' .$$

b) Iterate this equation a few times and show that your result agrees with the exact result expanded to the appropriate order.

14. Show that $x_1(t)$ as given by Eq. (8.49) satisfies the requirement that energy be conserved *through order* δ where $\delta = \epsilon x_0/\omega_0^2 \equiv ax_0/k$. That is, show that

$$\tfrac{1}{2}m\dot{x}_1^2 + \tfrac{1}{2}kx_1^2 + \tfrac{1}{3}ax_1^3 = \tfrac{1}{2}kx_0^2 + \tfrac{1}{3}ax_0^3 + \mathcal{O}(\delta^2) ,$$

where $\mathcal{O}(\delta^2)$ denotes terms of order δ^2 and higher.

15. In solving the equation for the anharmonic oscillator in Section 8.4, we used the notation $x_0(t)$, $x_1(t)$, $x_2(t)$, etc., to denote the zeroth iterate, first iterate, second iterate, and so forth. Similarly, let us denote by $\bar{x}_0(t)$, $\bar{x}_1(t)$, $\bar{x}_2(t)$, etc., that part of the iterative solution containing all terms through order δ^0, order δ^1, order δ^2, and so forth. Clearly, $\bar{x}_0(t) = x_0(t)$ and $\bar{x}_1(t) = x_1(t)$, but $\bar{x}_2(t)$ contains only part of $x_2(t)$. Find $\bar{x}_3(t)$ [this will involve computing only a few of the terms in $x_3(t)$] and show that all terms in $\bar{x}_3(t)$ which are *linear* in t may be removed, *through order* δ^3, by replacing ω_0 by ω_1 in all the trigonometric functions. Write ω_1 as

$$\omega_1 = \omega_0[1 + a\delta + b\delta^2 + c\delta^3]^{1/2} .$$

Find the values of a, b, and c which remove the terms linear in t. Using physical arguments, determine how large x_0 could be before the above expansion for ω_1 would diverge (we are imagining that the above redefinition of the frequency is carried out to all orders in δ).

16. a) Obtain an integral equation equivalent to the differential equation for the anharmonic oscillator whose motion is governed by

$$\ddot{x}(t) + \omega_0^2 x(t) = -ax^3(t) ,$$

by making use of the Green's function for the operator $d^2/dt^2 + \omega_0^2$. Assume $x(0) = x_0$, $\dot{x}(0) = 0$.

b) Iterate this equation once to obtain the first correction to the lowest-order solution, $x_0(t) = x_0 \cos \omega_0 t$.

17. For the system of Problem 8.16, find $\bar{x}_2(t)$ (we use the notation introduced in Problem 8.15). Show that if we write

$$\omega_1 = \omega_0 [1 + a\delta + b\delta^2]^{1/2},$$

then by appropriate choice of a and b, we can eliminate *through order* δ^2 all terms in $\bar{x}_2(t)$ proportional to t and t^2 by replacing ω_0 by ω_1 in all the trigonometric functions.

18. Solve the equations

a) $f(x) = x + \lambda \int_0^\infty e^{-(x+y)} f(y)\, dy,$ b) $f(x) = x + \frac{\lambda}{\pi} \int_0^{\pi/2} \cos(x + y) f(y)\, dy.$

19. Find the unique bound-state solution to Schrödinger's integral equation for a separable potential of the form $-g^2 v(r) v(r')$ with $v(r) = e^{-\mu r}/r$. Show that a bound-state solution exists if and only if $g^2 > \hbar^2\mu^3/(4\pi m)$, where m is the mass of the bound particle. Show also that the energy of the bound state is

$$E_B = -\frac{\hbar^2}{2m}\left[\left(\frac{4\pi m g^2}{\mu\hbar^2}\right)^{1/2} - \mu\right]^2.$$

What is the bound-state wave function?

20. Consider the quantity

$$f^S(E) = \frac{2\tau}{(q_i^2 + \mu^2)^2 + \frac{\tau}{\mu}(q_i - i\mu)^2},$$

where $\tau = 4\pi m g^2/\hbar^2$ and $q_i = (2mE/\hbar^2)^{1/2}$ [see Eq. (8.73)]. Show that if we consider E to be a complex variable, that is, if we do not restrict E to take on just physical real values, then $f^S(E)$ is a single-valued function of E on a two-sheeted Riemann surface cut along the positive real axis. Show that, as a function of E, $f^S(E)$ is analytic everywhere except for a pole of second order at $E = -\hbar^2\mu^2/2m$ on the first sheet, a simple pole at $E = -\hbar^2(\mu + \sqrt{\tau/\mu})^2/2m$ on the second sheet, and a so-called bound-state pole which may be either on the first or second sheet. If $\mu > \sqrt{\tau/\mu}$, then this pole is on the second sheet at $E = -\hbar^2(\sqrt{\tau/\mu} - \mu)^2/2m$; if $\sqrt{\tau/\mu} > \mu$, the pole is on the first sheet at $E = -\hbar^2(\sqrt{\tau/\mu} - \mu)^2/2m$. This value of E is precisely the energy of the bound state determined in the previous problem. We see, therefore, that as g^2 drops below the critical value necessary to produce a bound state, the effect of the bound state does not magically vanish from the scattering amplitude; the pole corresponding to the bound state merely moves from the first ("physical") sheet to the second sheet.

21. At the beginning of Section 8.6 in discussing kernels of finite rank, we introduced briefly the notion of a linear functional. The purpose of this problem and the two that follow it is to introduce some of the simpler aspects of linear functionals on a Banach space. We make the following *definition*. A *functional* F, on a Banach space B, is an operation which assigns to every $g \in B$ a complex number (assuming that B is defined over the field of complex numbers). The action of F on g is denoted by $F(g)$. F is said to be *linear* if $F(\alpha g_1 + \beta g_2) = \alpha F(g_1) + \beta F(g_2)$ for all complex numbers, α and β, and all $g_1, g_2 \in B$. F is said to be *bounded* if there exists a real number M, such that $|F(g)| \leq M||g||$ for all $g \in B$, M being independent of g. Let us now consider two aspects of this.

a) Let F be a bounded, linear functional on B. Suppose that we have in B a sequence $\{f_n\}$ which tends in norm to $f \in B$. Show that $F(f_n) \to F(f)$, that is, that F is continuous.

b) Show that if $f \in L_p$, then the functional F defined on L_q $(1/p + 1/q = 1)$ by

$$F(g) \equiv (f, g) \equiv \int f^*(x) g(x)\, dx\,,$$

where $g \in L_q$, is a bounded, linear functional. [*Hint*: Look back at Problem 8.2]. It can be shown that every bounded, linear functional on L_q can be written in this form, i. e., can be generated by an element of L_p.

22. Just as for a bounded, linear operator on B, we define the norm of a bounded, linear functional on B as the least uppper bound of the set of all M satisfying $|F(g)| \leq M||g||$ for all $g \in B$. As usual, we denote the norm of F by $||F||$.

a) Show that for the functional F, defined on L_q by

$$F(g) = (f, g)\,,$$

where $f \in L_p$ and $1/p + 1/q = 1$ (see the preceding problem), we have $||F|| \leq ||f||$. Here the norm of f is the norm appropriate to L_p, namely,

$$||f|| = \left[\int |f(x)|^p\, dx\right]^{1/p}.$$

b) Show that the function

$$g = \frac{|f|^{(p/q)+1}}{f^*}$$

belongs to L_q if $f \in L_p$ $(1/p + 1/q = 1)$ and therefore that $||F|| \geq ||f||$. Do this by showing that

i) $|F(g)| = |(f, g)| = ||f||^p$, ii) $||f||\,||g|| = ||f||^p$.

Hence $|F(g)| = ||f|| \cdot ||g||$, so $||F|| \geq ||f||$. Combining parts (a) and (b), we see that actually $||F|| = ||f||$.

23. Show that, by an obvious definition of addition and scalar multiplication, the bounded linear *functionals* on B form a Banach space in their own right. This space is called the *dual space* of B and is denoted by B^*. Since every linear functional on L_q is generated by an element of L_p $(1/p + 1/q = 1)$, we can identify L_q^* with L_p and say that L_p is the *dual space* of L_q. Note that the space L_2 of square-integrable functions is its own dual space.

There are three parts to this problem. First, you must show that your definition of addition and scalar multiplication is such that it gives rise to a linear vector space. Second, you must show that the definition of norm given in the previous problem really is a norm (as we did in Section 8.2 in discussing norms of operators). These two parts are easy. Not so easy is the problem of showing that the normed, linear vector space which you have constructed is *complete*.

[*Hint*: You must show that any Cauchy sequence of functionals $\{F_n\}$ has a limit which is also a bounded linear functional. Note that for any $f \in B$, $\{F_n(f)\}$ is a Cauchy sequence of complex numbers. Since the complex numbers are complete, you can define the limit of $\{F_n\}$ acting on any $f \in B$ in terms of the limit of $\{F_n(f)\}$, which is guaranteed to exist. Denote this limit by $F(f)$. It is then not difficult to

show that F so defined is linear and bounded and that $||F_n - F|| \to 0$ as $n \to \infty$. The basic idea for all these proofs is contained in the following identity which would be relevant for the proof of the linearity of F:

$$F(\alpha g_1 + \beta g_2) - [\alpha F(g_1) + \beta F(g_2)] \equiv [F(\alpha g_1 + \beta g_2) - F_n(\alpha g_1 + \beta g_2)] - [\alpha F(g_1) - \alpha F_n(g_1)] - [\beta F(g_2) - \beta F_n(g_2)] .$$

Here we make use of the *known* linearity of F_n.]

24. Show that by defining addition and scalar multiplication in the obvious way, we can construct a Banach space out of the set of all bounded, linear *operators* on B. This proof can be patterned directly on the proof given in the previous problem for bounded linear functionals. Again, the completeness of the underlying space B, is the key ingredient in the proof.
25. Using Eqs. (8.75) and (8.88), show explicitly that

$$(I - \lambda K_F)(I + \lambda R_{K_F}) = I .$$

26. Show that if K_N is defined by Eq. (8.93) and P_N by Eq. (8.94), then $K_N = P_N K P_N$.

FURTHER READINGS

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CHAPTER 9

INTEGRAL EQUATIONS IN HILBERT SPACE

INTRODUCTION

In the last chapter we developed two of the basic techniques for the solution of integral equations, iteration and finite-rank approximation, and in this chapter we wish to exploit these methods to obtain some results which facilitate greatly the solution of practical problems. We shall begin by looking at the properties of completely continuous Hermitian operators in some detail; in particular, we shall obtain a completeness relation for the eigenfunctions of such operators and use these eigenfunctions to discuss perturbation theory and the solution of linear equations.

With the insights obtained from this special case, we will move on to consider general completely continuous operators and see how to solve linear equations by the method of finite-rank approximation. Finally, we will close the chapter by looking at the Fourier transform in the context of the theory of unitary operators on Hilbert space. This is one of the most important tools in mathematical physics, and the ease with which it can be handled in the context of infinite-dimensional vector spaces is a beautiful illustration of the power of the theory of linear operators.

9.1 COMPLETELY CONTINUOUS HERMITIAN OPERATORS

Before we attempt to deal with the most general type of completely continuous operators, we will investigate an important special case: the completely continuous, self-adjoint operators. This kind of operator is interesting in its own right; for example, the bound-state Schrödinger integral equations can be handled very deftly using the techniques we shall develop here. But in addition, this special case will give us some important clues as to how we can proceed to treat the general theory of completely continuous operators.

Just as in the finite-dimensional case, a completely continuous self-adjoint operator A satisfies

$$(x, Ay) = (Ax, y) \tag{9.1}$$

for all x, y in the vector space. Since A is completely continuous, it is also bounded, and therefore is defined on every element of the space. If A is an integral operator with kernel $a(t, t')$, then the Hermiticity requirement is equivalent to

$$a(t, t') = a^*(t', t) , \tag{9.2}$$

since

$$(x, Ay) = \int dt \int dt' \, x^*(t) a(t, t') y(t') \, ,$$

$$(Ax, y) = \int dt' \int dt \, a^*(t, t') x^*(t') y(t) = \int dt' \int dt \, x^*(t) a^*(t', t) y(t') \, .$$

Evidently Eq. (9.1) holds if Eq. (9.2) is obeyed. It should be said at this point that our change in notation, with the operator written as A and the vectors as x and y, is not capricious. We shall see in the next two sections that the theory of completely continuous, Hermitian operators is virtually identical to the finite-dimensional theory we discussed in Chapter 4. To emphasize this similarity, we shall observe the notational conventions of Chapter 4 in this section and the next. However, the reader should remember that the vectors x and y may be functions in this chapter, in which case we will write them as $x(t)$ and $y(t)$ if it is necessary to display their functional dependence.

We begin by recalling the definition of norm which we gave in Section 8.2, namely, $||A||$ is the smallest constant such that

$$||Ax|| \leq ||A|| \cdot ||x||$$

for all $x \in B$, where B is any normed vector space and A is a bounded operator. Now it is interesting to inquire whether or not there is actually some vector in the space for which one has

$$Ay = \pm ||A|| y \, . \tag{9.3}$$

Let H_1 be a Hilbert space, and let A be an operator with the property that there is always some vector in the space which satisfies Eq. (9.3). Suppose that we have found some vector x_1 which satisfies

$$Ax_1 = \pm ||A||_1 x_1 \, ,$$

where $||A||_1$ is the norm of A when it acts on the whole Hilbert space, H_1. The space of all vectors orthogonal to x_1, say, H_2, is also a Hilbert space; if A were bounded on H_1, then A would be necessarily bounded on H_2. Thus

$$||Ax|| \leq ||A||_2 \cdot ||x||$$

for all $x \in H_2$. Clearly, $||A||_2 \leq ||A||_1$. Now if A still has the property assumed above, then there is some $x_2 \in H_2$ such that

$$Ax_2 = \pm ||A||_2 x_2 \, ,$$

where $(x_1, x_2) = 0$. We can continue this process indefinitely, generating H_1, H_2, H_3, $\cdots$ and x_1, x_2, x_3, $\cdots$.

Several possibilities may now arise. It might be that $||A||_1 = ||A||_2 = ||A||_3 = \cdots$. This would be the case, for instance, if A were the identity operator. However, if A is completely continuous and $A \neq 0$, then this cannot happen, i.e., we cannot obtain some nonzero eigenvalue an infinite number of times. For if we did, we would have an infinite orthonormal sequence $\{x_n\}$,

and $\{Ax_n\}$ would need a convergent subsequence since A is completely continuous. But if $Ax_n = ax_n$ for all the vectors of the sequence, then $\{ax_n\}$ must contain a convergent subsequence. However, $||x_m - x_n|| = \sqrt{2}$ for all m, n, so there can be no convergent subsequence, unless, of course, $a = 0$. Hence we can obtain a given, nonzero eigenvalue only a finite number of times.

However, it is possible to obtain a zero eigenvalue an infinite number of times. This, for example, would be true for an operator of finite rank on a Hilbert space. In any case, however, the eigenvalues λ_n must tend toward zero, because if they did not, x_n/λ_n would be a bounded sequence. But $Ax_n = \lambda_n x_n$, so $\{x_n\}$ must contain a convergent subsequence, which it cannot as noted above. This behavior of the eigenvalues could arise either because all but a finite number of eigenvalues are zero, or because there is an infinite collection of nonzero eigenvalues, λ_n, all of finite multiplicity, which become arbitrarily small for n sufficiently large. In this latter case, we might also have some zero eigenvalues (perhaps even an infinite number of them). Corresponding to these eigenvalues, we have an infinite number of eigenvectors, x_n, and we may hope that they will form a complete set, although the fact that there is an infinite number of x_n is only necessary for completeness, not sufficient.

In finite-dimensional spaces, the completeness property held for the eigenfunctions of self-adjoint, isometric, and normal transformations. In the case of completely continuous self-adjoint transformations, we shall now show that this same result still holds. Of course, basic properties such as reality of eigenvalues will be the same since the proofs of these properties given in Chapter 4 were in no way restricted to the finite-dimensional case. However, by way of emphasizing the difference between the finite-dimensional case and the completely continuous case, we mention that there are *no* completely continuous *isometric* transformations of an infinite-dimensional Hilbert space. Clearly, every isometry is *bounded* since

$$||Uy||^2 = (Uy, Uy) = (y, y) = ||y||^2$$

for any $y \in H$. Thus, $||U|| = 1$. However, consider any sequence $\{y_n\}$, which does not contain a convergent subsequence, and the related sequence $\{Uy_n\}$;

$$||Uy_n - Uy_m|| = ||U(y_n - y_m)|| = ||y_n - y_m|| ,$$

so U does not change the convergence properties of $\{y_n\}$ in any way. In particular, if $\{y_n\}$ does not contain a convergent subsequence, neither will $\{Uy_n\}$. Thus isometric transformations are not completely continuous.

The most important property of self-adjoint transformations is that they have *real* expectation values. This fact leads us to the following important result.

Theorem 9.1. Let A be a bounded, self-adjoint linear operator on the Hilbert space, H. Then if $||y|| = 1$,

$$\sup_{y \in H} |(y, Ay)| = ||A|| ,$$

where sup denotes the least upper bound.

Proof. By the Schwartz inequality, for all $y \in H$,

$$|(y, Ay)| \leq ||y|| \cdot ||Ay|| \leq ||A|| \cdot ||y||^2 . \tag{9.4}$$

Now, in analogy with our definition of the norm of A, we define the quantity $|A|$ as the smallest constant for which

$$|(y, Ay)| \leq |A| \cdot ||y||^2 , \tag{9.5}$$

for all $y \in H$. According to Eq. (9.4), $|A| \leq ||A||$. We now show that $|A| \geq ||A||$. To do this, we utilize the identity

$$||Ay||^2 = (Ay, Ay) \equiv \frac{1}{4}\left\{\left(\lambda y + \frac{1}{\lambda} Ay, A\left[\lambda y + \frac{1}{\lambda} Ay\right]\right) - \left(\lambda y - \frac{1}{\lambda} Ay, A\left[\lambda y - \frac{1}{\lambda} Ay\right]\right)\right\} ,$$

which holds for all real λ. Thus, using Eq. (9.5), we get

$$||Ay||^2 \leq \frac{1}{4} |A| \cdot \left[||\lambda y + \frac{1}{\lambda} Ay||^2 + ||\lambda y - \frac{1}{\lambda} Ay||^2 \right] ,$$

so that

$$||Ay||^2 \leq \frac{1}{2} |A| \cdot \left(\lambda^2 \cdot ||y||^2 + \frac{1}{\lambda^2} ||Ay||^2 \right) .$$

In particular, the inequality holds if $\lambda^2 = ||Ay||/||y||$. From this value, we find that

$$||Ay||^2 \leq |A| \cdot ||Ay|| \cdot ||y|| , \qquad \text{or} \qquad ||Ay|| \leq |A| \cdot ||y|| .$$

Thus $||A|| \leq |A|$. Since we have already shown that $||A|| \geq |A|$, we see that $||A|| = |A|$, as required. Note that this result holds for any bounded, self-adjoint operator, regardless of whether or not it is completely continuous. It might almost appear not to even depend on the Hermiticity of A. However, the reader can readily show that the identity in λ which is crucial to the proof is only true if A is self-adjoint.

With this result in hand, we can now carry out the program outlined at the beginning of this section.

Theorem 9.2. Let A be a completely continuous, self-adjoint transformation on a Hilbert space, H, and denote by C the set of all $y \in H$ such that $||y|| = 1$. The quantity $|(y, Ay)|$ attains a maximum value on this set, and the vector x, for which the maximum is attained is an eigenvector of A with eigenvalue $\lambda = \pm ||A||$.

Proof. Choose a sequence $\{y_n\}$, $y_n \in C$, such that (y_n, Ay_n) tends to λ_1, where $|\lambda_1| = ||A||$. The previous theorem tells us that this is always possible. Since we want to prove that λ_1 is an eigenvalue, let us consider

$$||Ay_n - \lambda_1 y_n||^2 = ||Ay_n||^2 - 2\lambda_1 (y_n, Ay_n) + \lambda_1^2 ||y_n||^2 .$$

Now $||Ay_n||^2 \leq ||A||^2 \cdot ||y_n||^2 = \lambda_1^2 ||y_n||^2$, so since $||y_n|| = 1$,

$$||Ay_n - \lambda_1 y_n||^2 \leq 2[\lambda_1^2 - \lambda_1(y_n, Ay_n)] .$$

But since $(y_n, Ay_n) \to \lambda_1$ by assumption,

$$||Ay_n - \lambda_1 y_n|| \to 0 . \tag{9.6}$$

Since A is completely continuous, the sequence $\{Ay_n\}$ contains a convergent subsequence, say, $\{A\eta_m\}$, where $\{\eta_m\}$ is a subset of $\{y_n\}$. For reasons which will soon be clear, we denote the vector to which $\{A\eta_m\}$ converges by $\lambda_1 x_1$:

$$||A\eta_m - \lambda_1 x_1|| \to 0 . \tag{9.7}$$

But

$$A\eta_m - \lambda_1 \eta_m \equiv A\eta_m - \lambda_1 x_1 + \lambda_1 x_1 - \lambda_1 \eta_m , \tag{9.8}$$

so

$$||\lambda_1 x_1 - \lambda_1 \eta_m|| \leq ||A\eta_m - \lambda_1 x_1|| + ||A\eta_m - \lambda_1 \eta_m|| .$$

Thus by Eqs. (9.6) and (9.7),

$$||\lambda_1 x_1 - \lambda_1 \eta_m|| \to 0; \tag{9.9}$$

in other words, the sequence $\{\eta_m\}$ converges to x_1 (assuming, of course, that $|\lambda_1| = ||A|| \neq 0$). By the triangle inequality,

$$\begin{aligned} ||Ax_1 - \lambda_1 x_1|| &\leq ||Ax_1 - A\eta_m|| + ||A\eta_m - \lambda_1 \eta_m|| + ||\lambda_1 \eta_m - \lambda_1 x_1|| \\ &\leq 2||A|| \cdot ||x_1 - \eta_m|| + ||A\eta_m - \lambda_1 \eta_m|| , \end{aligned}$$

so by Eqs. (9.6) and (9.9), we can make

$$||Ax_1 - \lambda_1 x_1|| < \epsilon$$

for any ϵ. Hence

$$Ax_1 = \lambda_1 x_1 ,$$

so that λ_1 is an eigenvalue of A corresponding to the eigenvector x_1. Also,

$$|(x_1, Ax_1)| = |(x_1, \lambda_1 x_1)| = |\lambda_1| = ||A|| .$$

This completes the proof of this theorem, which is the basic result on completely continuous, self-adjoint operators.

Now, let us consider again the Hilbert space H_2 of all vectors orthogonal to the vector x_1 found in the previous theorem. The operator A acting on this space is completely continuous and Hermitian. We write

$$||Ay|| \leq ||A||_2 \cdot ||y||$$

for all $y \in H_2$, where $||A||_2 \leq ||A||$. By repeating the procedure of the above theorem, we can find an x_2 such that

$$Ax_2 = \lambda_2 x_2 , \qquad |\lambda_2| = ||A||_2 .$$

We can continue the process indefinitely unless we arrive at a number, m, where

the eigenvalue $\lambda_m = 0$ that is, $||A||_m = 0$. This means that after removing a number, m, of orthonormal vectors from our original space, H, we arrive at a space H_{m+1} such that A acting on this space produces only zero vectors. This means that H_{m+1} is the *null space* of A. This will clearly be the case for the operator of finite rank, A_F, defined by

$$A_F y \equiv \sum_{n=1}^{N} \alpha_n \phi_n (\phi_n, y) ,$$

where the ϕ_n form an orthonormal set. A_F has as eigenvectors the ϕ_n with corresponding eigenvalues, α_n. A_F also has as eigenvectors any vectors which are orthogonal to all the ϕ_n. Zero is the common eigenvalue for all these other eigenvectors. We may summarize the above discussion as follows:

Theorem 9.3. Let Y be the set of all vectors $y \in H$ such that $||y|| = 1$ and y is orthogonal to all the vectors $x_1, x_2, \cdots, x_n$ satisfying

$$Ax_n = \lambda_n x_n .$$

Then $|(y, Ay)|$ can always be maximized on this set if A is a completely continuous, self-adjoint operator. Furthermore, the vector x_{n+1} obtained in this way is an eigenvector of A with eigenvalue λ_{n+1}, where $|\lambda_{n+1}| = ||A||_{n+1}$ and $||A||_{n+1}$ is the norm of A in the subspace of H consisting of all vectors orthogonal to $x_1, x_2, \cdots, x_n$. This process generates an infinite sequence of vectors $\{x_n\}$ and a set of eigenvalues $\{\lambda_n\}$. Each nonzero eigenvalue is of finite multiplicity and $\lambda_n \to 0$ as $n \to \infty$.

Now we are in a position to obtain a completeness theorem for completely continuous, self-adjoint operators on a Hilbert space.

Theorem 9.4. The eigenvectors of the completely continuous, self-adjoint operator A form a complete, orthonormal set of vectors in the Hilbert space, H, on which A acts.

Proof. Let us assume at first that A has only nonzero eigenvalues.* It is clear that at the very least there is always a subspace of H in which this is true. For any $y \in H$, we consider

$$y_N = y - \sum_{n=1}^{N} (x_n, y) x_n ,$$

where the set $\{x_n\}$ consists of the eigenvectors of A. To establish completeness, we must show that as $N \to \infty$, $||y_N|| \to 0$. But y_N is obviously orthogonal to all x_n with $n \leq N$. Hence by our previous work

$$||Ay_N|| \leq |\lambda_{N+1}| \cdot ||y_N|| .$$

* We will also assume that A has an infinite number of eigenvectors belonging to nonzero eigenvalues. The case when A has only a finite number of nonzero eigenvalues is elementary and is left as a problem.

Since $||y_N|| \leq ||y||$, we have

$$||Ay_N|| \leq |\lambda_{N+1}| \cdot ||y|| .$$

But $\lambda_{N+1} \to 0$ as $N \to \infty$, so

$$||Ay_N|| \to 0$$

as $N \to \infty$. But it is easily shown that

$$Ay_N = Ay - \sum_{n=1}^{N} (x_n, y) Ax_n = Ay - \sum_{n=1}^{N} (x_n, Ay) x_n .$$

Thus we may say that

$$\sum_{n=1}^{N} (x_n, Ay) x_n$$

tends in norm to Ay as $N \to \infty$. We write this as

$$Ay \doteq \sum_{n=1}^{\infty} (x_n, Ay) x_n , \tag{9.10}$$

for any $y \in H$. Hence we have a completeness relation for all elements of the space of the form Ay.

If we knew that A were invertible, then we could immediately extend the completeness relation to all vectors in H, since then any vector in H could be written in the form Ay for some $y \in H$. However, more generally, we may argue as follows. Since H is a complete space,

$$\lim_{N \to \infty} \left[\sum_{n=1}^{N} (x_n, \eta) x_n \right]$$

converges in the mean to some vector, say $\bar{\eta}$, because Bessel's inequality (Eq. 5.9) tells us that the sequence

$$\left\{ \eta_N = \sum_{n=1}^{N} (x_n, \eta) x_n \right\}$$

is a *Cauchy* sequence. We write

$$\bar{\eta} \doteq \sum_{n=1}^{\infty} (x_n, \eta) x_n . \tag{9.11}$$

Since A is continuous,

$$A\bar{\eta} \doteq \sum_{n=1}^{\infty} (x_n, \eta) Ax_n \doteq \sum_{n=1}^{\infty} (x_n, A\eta) x_n .$$

But by Eq. (9.10),

$$A\eta \doteq \sum_{n=1}^{\infty} (x_n, A\eta) x_n .$$

Hence

$$A\tilde{\eta} - A\eta = A(\tilde{\eta} - \eta) = 0 ,$$

except possibly on a set of measure zero. But we have assumed that A has no zero eigenvalues, so

$$\tilde{\eta} = \eta .$$

Thus Eq. (9.11) becomes

$$\eta \doteq \sum_{n=1}^{\infty} (x_n, \eta) x_n \tag{9.12}$$

for any $\eta \in H$, which is the familiar completeness relation. It should be remembered that the convergence of the infinite sum in Eq. (9.12) is defined in the sense of *mean* convergence.

The case when $Ax = 0$ has solutions is easily handled. The set of all such solutions forms a Hilbert space, and we may pick an orthonormal basis $\{y_n\}$ for it. (Note that according to Theorem 4.18, every y_n is orthogonal to every x_m belonging to a non-zero eigenvalue.) Each element of the basis satisfies $Ay_n = 0$. Now given any x, form

$$\tilde{x} = x - \sum_{n=1}^{\nu} (y_n, x) y_n ,$$

where ν is the number of elements in the set $\{y_n\}$. Clearly, $\tilde{x}$ lies in the orthogonal complement of the space spanned by $\{y_n\}$, so we can apply the above result to it:

$$\tilde{x} \doteq \sum_{m=1}^{\infty} (x_m, \tilde{x}) x_m \doteq \sum_{m=1}^{\infty} (x_m, x) x_m ,$$

where the $\{x_m\}$ are the eigenvectors of A when A is restricted to act on the orthogonal complement of the space spanned by the y_n. Hence

$$x \doteq \sum_{n=1}^{\nu} (y_n, x) y_n + \sum_{m=1}^{\infty} (x_m, x) x_m$$

for any $x \in H$. Therefore the set $\{x_n, y_m\}$ is a complete orthonormal basis in H, and we are done.

As a simple example of some of the previous results consider the equation

$$\int_0^1 t_<(1 - t_>) x_n(t') \, dt' = \lambda_n x_n(t), \tag{9.13}$$

which was shown in Section 7.1 to be equivalent to the differential eigenvalue problem

$$\frac{d^2 x_n(t)}{dt^2} + \mu_n x_n(t) = 0 , \qquad x_n(0) = 0 = x_n(1) ,$$

where $\mu_n = 1/\lambda_n$. The eigenvectors of Eq. (9.13) are therefore just

$$x_n(t) = \sqrt{2}\sin(n\pi t) ,$$

with eigenvalues

$$\lambda_n = 1/n^2\pi^2 .$$

This may also be verified by direct substitution in Eq. (9.13). Theorem 9.4 tells us that the vectors $x_n(t) = \sqrt{2}\sin(n\pi t)$ are a complete set of orthonormal functions on the space of square-integrable functions on [0, 1] which vanish at $t = 0$ and $t = 1$. From our previous discussion, we know that the norm of the integral operator defined by Eq. (9.13) must be equal to the largest eigenvalue, $1/\pi^2$. We have also seen that the norm of an integral kernel is less than or equal to the square root of the integrated square of the kernel [see, for example, Eq. (8.47) et. seq.]. In this case, therefore, we must have

$$\lambda_1^2 \leq \int_0^1 dt \int_0^1 dt'\, t_<^2(1 - t_>)^2 = 1/90 \qquad \text{or} \qquad |\lambda_1| \leq 1/9.49 = 0.1054 .$$

The actual value of λ_1 is $1/\pi^2$, that is, $\lambda_1 = 0.1013$, so we see that by approximating the norm of A by the square root of the integrated square of its kernel, we get an excellent estimate of the lowest eigenvalue. We can also estimate the second eigenvalue in this way by calculating in this approximate manner the norm of the operator *restricted* to the space of all vectors orthogonal to x_1. This means calculating the norm of

$$(I - P_1)A(I - P_1) ,$$

where P_1 is the projection onto the subspace spanned by x_1. In terms of the kernel of A, $a(t, t') = t_<(1 - t_>)$; this means that

$$|\lambda_2| \leq \left[\int_0^1 dt \int_0^1 dt' [a(t, t') - \lambda_1 x_1(t')]^2 \right]^{1/2} .$$

Since x_1 is an eigenvector of A, this integral reduces immediately to

$$|\lambda_2| \leq \left[\int_0^1 dt \int_0^1 dt' (a(t, t')^2 - \lambda_1^2) \right]^{1/2} .$$

Both quantities in the bracket are known, and we find that

$$|\lambda_2| \leq 0.029 ,$$

which is quite close to the exact value, $\lambda_2 = 1/4\pi^2 = 0.025$. For λ_1, the approximation was good to 4%, while for λ_2 the result is accurate to only 16%. Applying the some method to λ_3 yields

$$|\lambda_3| \leq 0.015 ,$$

compared with the exact value, $\lambda_3 = 1/9\pi^2 = 0.011$, which tells us that the estimate continues to worsen as we go to higher eigenvalues. However, such estimates are often useful in practical problems in getting an idea of the size of

the lowest eigenvalue. Clearly, the method cannot be used to estimate higher eigenvalues unless the lower eigenvalues are known *exactly*.

These estimates can be understood more easily if we consider the following corollary of the previous theorem.

Corollary. If A is a self-adjoint integral operator whose square-integrable kernel is $a(t, t')$, then

$$a(t, t') \doteq \sum_{n=1}^{\infty} \lambda_n x_n(t) x_n^*(t') , \tag{9.14}$$

where $\{x_n\}$ and $\{\lambda_n\}$ are respectively the eigenvectors and eigenvalues of A, and the infinite sum is defined in the sense of mean convergence.

Proof. Denote the sum by

$$S(t, t') \doteq \sum_{n=1}^{\infty} \lambda_n x_n(t) x_n^*(t') .$$

We leave it for the reader to show that this sum is indeed convergent; this is just a matter of satisfying the Cauchy condition. Define the difference function

$$d(t, t') \equiv a(t, t') - S(t, t') ,$$

and consider the related integral operator, D:

$$Dy = \int d(t, t') y(t') \, dt' .$$

Let us assume that $||D||$ is *not* equal to zero. Now by Theorem 9.2, D has at least one nonzero eigenvalue, say λ, with associated eigenvector, $x(t)$. Let $x_n(t)$ $(n = 1, 2, \cdots)$ be any eigenvector of A [and hence of S, where S is the integral operator whose kernel is $S(t, t')$]. Consider

$$(x, x_n) \equiv \frac{1}{\lambda^*} (\lambda x, x_n) = \frac{1}{\lambda^*} (Dx, x_n) = \frac{1}{\lambda^*} (x, Dx_n) .$$

But

$$Dx_n = \int [a(t, t') - s(t, t')] x_n(t') \, dt' = (A - S) x_n = 0 .$$

Hence

$$(x, x_n) = 0 .$$

Thus, we have found a nonzero vector which cannot be expanded in terms of the supposedly complete set $\{x_n\}$, which contradicts the completeness theorem (Theorem 9.4). Hence our assumption that $||D|| \neq 0$ was incorrect, and therefore $||D|| = 0$. Thus, $d(t, t') = 0$, which proves the corollary.

This tells us that *every* symmetric, square-integrable function of two variables can be expanded in a series of the form of Eq. (9.14). We also have the

result, which we leave to the reader to show, that

$$\int dt \int dt' \cdot |a(t, t')|^2 = \sum_{n=1}^{\infty} \lambda_n^2 , \tag{9.15}$$

where $\{\lambda_n\}$ is the set of eigenvalues of the integral operator A whose kernel is given by the symmetric function $a(t, t')$. Equation (9.15) is the basis of our estimates of $\lambda_1, \lambda_2, \cdots$ given above. It tells us that if there are no eigenvalues "near" λ_1, that is, $\lambda_1^2 \gg \lambda_2^2 > \lambda_3^2 > \cdots$, then

$$\int dt \int dt' \, |a(t, t')|^2 \cong \lambda_1^2 .$$

Until now we have ordered the eigenvalues according to decreasing magnitude of $|\lambda_n|$, $|\lambda_1|$ being the maximum value of $|(y, Ay)|$ which is attained when $y = x_1$, where

$$Ax_1 = \lambda_1 \cdot x_1 .$$

λ_1 may be positive or negative. The previous theorem enables us to make a similar ordering for the positive and negative eigenvalues separately.

Theorem 9.5. Suppose that A is a completely continuous Hermitian operator. Let $\lambda_1^+, \lambda_2^+, \cdots$ be the positive eigenvalues of A ordered so that $\lambda_1^+ \geq \lambda_2^+ \geq \cdots$, with corresponding eigenvectors $x_1^+, x_2^+, \cdots$. Similarly, let $\lambda_1^-, \lambda_2^-, \cdots$ be the negative eigenvalues of A with $\lambda_1^- \leq \lambda_2^-, \leq \cdots$, and denote the corresponding eigenvectors by $x_1^-, x_2^-, \cdots$. Then λ_{n+1}^+ is the *maximum* value of (y, Ay) when y varies over the Hilbert space, H, subject to $||y|| = 1$ and $(y, x_i^+) = 0$ for $i = 1, 2, \cdots, n$. Similarly, λ_{n+1}^- is the *minimum* value of (y, Ay) when y varies over H subject to $||y|| = 1$ and $(y, x_i^-) = 0$ for $i = 1, 2, \cdots, n$.

Proof. By Theorem 9.4, we have for any $y \in H$

$$Ay \doteq \sum_{n=1}^{\infty} (x_n, Ay) x_n \doteq \sum_{n=1}^{\infty} \lambda_n (x_n, y) x_n .$$

Therefore

$$(y, Ay) = \sum_{n=1}^{\infty} \lambda_n |(x_n, y)|^2 .$$

Using the notation just introduced, we can write this as

$$(y, Ay) = \sum_{\nu=1}^{N} \lambda_\nu^+ |(x_\nu^+, y)|^2 + \sum_{\mu=1}^{M} \lambda_\mu^- |(x_\mu^-, y)|^2 ,$$

where N is the number of positive eigenvalues and M is the number of negative eigenvalues (either M or N or both may be infinite). Suppose that $(y, x_i^+) = 0$ for $i = 1, 2, \cdots, n$. Then

$$(y, Ay) = \sum_{\nu=n+1}^{N} \lambda_\nu^+ |(x_\nu^+, y)|^2 + \sum_{\mu=1}^{M} \lambda_\mu^- |(x_\mu^-, y)|^2 .$$

Clearly, if $y = x_{n+1}^+$, then $(y, Ay) = \lambda_{n+1}^+$. Now consider any y and expand it as

$$y = \sum_{i=n+1}^{N} c_i^+ x_i^+ + \sum_{j=1}^{M} c_j^- x_j^- ,$$

where

$$\sum_{i=n+1}^{N} |c_i^+|^2 + \sum_{j=1}^{M} |c_j^-|^2 = 1; \tag{9.16}$$

y is orthogonal to $x_1^+, x_2^+, \cdots, x_n^+$ and is normalized to 1. For this y,

$$(y, Ay) = \sum_{\nu=n+1}^{N} \lambda_\nu^+ |c_\nu^+|^2 + \sum_{\mu=1}^{M} \lambda_\mu^- |c_\mu^-|^2 .$$

Since all λ_ν^+ with $\nu \geq n + 1$ satisfy $\lambda_\nu^+ \leq \lambda_{n+1}^+$, and since $\lambda_\mu^- \leq \lambda_{n+1}^+$ for all μ,

$$(y, Ay) \leq \lambda_{n+1}^+ \left[\sum_{\nu=n+1}^{N} |c_\nu^+|^2 + \sum_{\mu=1}^{M} |c_\mu^-|^2 \right] .$$

Thus according to Eq. (9.16)

$$(y, Ay) \leq \lambda_{n+1}^+ ,$$

which proves the result for positive eigenvalues. A similar argument gives the result for negative eigenvalues. We have *not* shown that $(y, Ay) = \lambda_{n+1}^+$ *only* for $y = x_{n+1}^+$; this would not be true, since λ_{n+1}^+ may have multiplicity $\mu > 1$. In our ordering of eigenvalues, we always list an eigenvalue and its corresponding eigenvector as many times as its multiplicity, μ, requires.

This theorem generalizes to infinite-dimensional spaces the results of Section 4.8 which were derived for finite-dimensional spaces. Thus we see that for completely continuous Hermitian operators, the results of finite-dimensional theory can be taken over to infinite-dimensional spaces virtually without alteration. Since we have obtained a complete, orthonormal set of eigenfunctions for A, it is also a straightforward matter to construct the projection operators, $E(\lambda)$, discussed in Section 5.11 and thereby write for any self-adjoint, completely continuous operator,

$$A = \int_{\lambda_{\min}}^{\lambda_{\max}} \lambda \, dE(\lambda) ,$$

where $E(\lambda)$ is the resolution of the identity belonging to A. Thus, for this special class of self-adjoint operators on Hilbert space we have essentially proved the Spectral Theorem. However, by no means are all self-adjoint operators on Hilbert space completely continuous. Even in the class of *bounded*, self-adjoint operators problems arise which prevent us from using the above methods to obtain the Spectral Theorem. Let us indicate briefly what one of the difficulties is.

Consider the bounded, Hermitian operator X, defined by

$$Xf(x) \equiv xf(x) ,$$

for all $f \in L_2[0, 1]$, where $L_2[0, 1]$ is the Hilbert space of square-integrable functions defined in the interval $[0, 1]$. Clearly, since $x \in [0, 1]$, the operator X is bounded. However, X is not completely continuous. Furthermore, the operator X has no eigenfunctions, since the equation

$$xf(x) = \lambda f(x)$$

cannot be solved to give any sensible sort of function. Of course, it is possible to find a nonsensible "eigenfunction," namely $\delta(x - x_0)$; that is, we could write

$$x\delta(x - x_0) = x_0\delta(x - x_0)$$

and say that $\delta(x - x_0)$ is an eigenfunction of X with eigenvalue x_0. This is true for any $x_0 \in [0, 1]$. Unfortunately, $\delta(x - x_0)$ is not a function in the usual sense of the word, and furthermore

$$\int_0^1 |\delta(x - x_0)|^2 \, dx = \infty$$

if $x_0 \in [0, 1]$, so $\delta(x - x_0)$ is not even square-integrable. However, the δ-function does suggest the following possibility. Consider any point $x_0 \in [0, 1]$. We can construct a sequence of functions, which in the sense of Eq. (9.6) may be called "approximate" eigenfunctions, as follows. We define

$$\phi_{x_0}^{(N)}(x) = \begin{cases} \sqrt{\dfrac{N}{2}}, & \text{if} \quad 0 \leq x_0 - \dfrac{1}{N} \leq x \leq x_0 + \dfrac{1}{N} \leq 1 , \\ 0 & \text{otherwise} ; \end{cases}$$

clearly $(\phi_{x_0}^{(N)}, \phi_{x_0}^{(N)}) = 1$. The *square* of $\phi_{x_0}^{(N)}$ looks like a δ-function when N is large. Now a simple calculation shows that

$$||X\phi_{x_0}^{(N)} - x_0\phi_{x_0}^{(N)}|| = ||(x - x_0)\phi_{x_0}^{(N)}|| = \frac{1}{3N^2} ,$$

so

$$||X\phi_{x_0}^{(N)} - x_0\phi_{x_0}^{(N)}|| \to 0$$

as $N \to \infty$. Thus the set $\{\phi_{x_0}^{(N)}\}$ provides us with an "approximate" eigenfunction whose eigenvalue is x_0, because given an ϵ, no matter how small, we can find an N such that $\phi_{x_0}^{(N)}$ satisfies

$$||X\phi_{x_0}^{(N)} - x_0\phi_{x_0}^{(N)}|| < \epsilon .$$

But unfortunately the sequence $\{\phi_{x_0}^{(N)}\}$ is not a Cauchy sequence, since

$$||\phi_{x_0}^{(N)} - \phi_{x_0}^{(M)}|| = 2(1 - \sqrt{M/N})$$

(where we suppose that $N \geq M$), and hence the sequence does not converge to

an element of $L_2[0, 1]$. We may summarize by saying that for every $x_0 \in [0, 1]$ we have an "approximate" eigenfunction with eigenvalue x_0, but there are *no* true eigenfunctions. For some operators it may even happen that there are both true and "approximate" eigenfunctions. It is because of difficulties like this that the concept of eigenfunctions and eigenvalues must be abandoned in order to obtain the Spectral Theorem for the most general type of self-adjoint operator.

9.2 LINEAR EQUATIONS AND PERTURBATION THEORY

Now that we have a completeness relation and a maximum-minimum principle available for completely continuous, self-adjoint operators, we can proceed to the important problems of perturbation theory and linear equations. To begin, let us consider the fundamental problem of solving a linear equation of the form

$$x = y + \lambda A x \,, \tag{9.17}$$

where A is a completely continuous, self-adjoint operator. Let $\{x_i\}$ denote the complete set of eigenvectors of A, with associated eigenvalues λ_i. Taking the inner product of both sides of Eq. (9.17) with x_i, we get

$$\begin{aligned}(x_i, x) &= (x_i, y) + \lambda(x_i, Ax) \\ &= (x_i, y) + \lambda\lambda_i(x_i, x) \,,\end{aligned}$$

and, therefore,

$$(1 - \lambda\lambda_i)(x_i, x) = (x_i, y) \,. \tag{9.18}$$

Thus if λ is not the reciprocal of any eigenvalue of A, then

$$(x_i, x) = \frac{(x_i, y)}{(1 - \lambda\lambda_i)} \,. \tag{9.19}$$

If $\lambda = \lambda_\nu^{-1}$, then Eq. (9.18) is well defined for $i = \nu$ only if $(x_\nu, y) = 0$, in which case the quantity (x_ν, x) is undetermined. If λ_ν has more than one associated eigenvector, a solution can exist only if y is orthogonal to all of them.

Now since

$$\phi \doteq \sum_{n=1}^{\infty} x_n(x_n, \phi)$$

in the sense of mean convergence, Eq. (9.19) tells us that

$$x \doteq \sum_{i=1}^{\infty} (1 - \lambda\lambda_i)^{-1}(x_i, y)x_i \,, \tag{9.20}$$

as long as λ is not the reciprocal of any λ_i. It is convenient to write this, using

$$(1 - \lambda\lambda_i)^{-1} = 1 + \lambda\lambda_i(1 - \lambda\lambda_i)^{-1} \,,$$

as

$$x \doteq \sum_{i=1}^{\infty} (x_i, y)x_i + \lambda \sum_{i=1}^{\infty} \frac{\lambda_i}{1 - \lambda\lambda_i} (x_i, y)x_i .$$

Hence

$$x \doteq y + \lambda R_A y , \tag{9.21}$$

where for any $\xi \in H$, R_A is defined by

$$R_A\xi \equiv \sum_{i=1}^{\infty} \frac{\lambda_i}{1 - \lambda\lambda_i} (x_i, \xi)x_i ; \tag{9.22}$$

R_A is called the *resolvent* operator of A. Note that R_A is self-adjoint and that the infinite sum in Eq. (9.22) converges in the mean, since $\lambda_i/(1 - \lambda\lambda_i)$ is bounded ($\lambda_i \to 0$, $\lambda \neq 1/\lambda_i$ for any i), and the sequence of partial sums is therefore a Cauchy sequence. In fact, we can even say that the operator sequence of partial sums, $\{R_A^{(N)}\}$, where

$$R_A^{(N)}\xi \equiv \sum_{i=1}^{N} \frac{\lambda_i}{1 - \lambda\lambda_i} (x_i, \xi)x_i ,$$

converges in operator norm to R_A, since

$$||(R_A - R_A^{(N)})\xi|| = \left[\sum_{i=N+1}^{\infty} \frac{\lambda_i^2}{(1 - \lambda\lambda_i)^2} |(x_i, \xi)|^2\right]^{1/2} \leq \frac{|\lambda_{N+1}|}{\Delta_{\min}} \left[\sum_{i=N+1}^{\infty} |(x_i, \xi)|^2\right]^{1/2};$$

$\Delta_{\min}$ is the minimum value of $|1 - \lambda\lambda_i|$ on the set of all eigenvalues of A, and we are ordering eigenvalues by decreasing absolute magnitude. Now

$$\sum_{i=N+1}^{\infty} |(x_i, \xi)|^2 \leq ||\xi|| ,$$

so

$$||(R_A - R_A^{(N)})\xi|| \leq \frac{|\lambda_{N+1}|}{\Delta_{\min}} ||\xi|| .$$

But $\lambda_{N+1} \to 0$ as $N \to \infty$, $||R_A - R_A^{(N)}|| \to 0$ as $N \to \infty$. This means that since $R_A^{(N)}$ is an operator of finite rank, and hence completely continuous, then by Theorem 8.3, R_A is also completely continuous. We may summarize as follows:

Theorem 9.6. Let A be a completely continuous, self-adjoint operator on a Hilbert space, H. Then if λ is not equal to the inverse of any eigenvalue of A, $(I - \lambda A)^{-1}$ exists and is given by

$$(I - \lambda A)^{-1} = I + \lambda R_A , \tag{9.23}$$

where R_A is a completely continuous Hermitian operator defined for any $\xi \in H$ by

$$R_A\xi \equiv \sum_{i=1}^{\infty} \frac{\lambda_i}{1 - \lambda\lambda_i} (x_i, \xi)x_i . \tag{9.24}$$

The infinite sum converges in the mean; $\{x_i\}$ is the set of all eigenvectors of A.

Now suppose that $\lambda = 1/\lambda_\nu$. Then according to our previous work,

$$x = \bar{x} + \sum_{i=1}^{\infty}{}' \frac{\lambda_i}{1 - \lambda\lambda_i} (x_i, y) x_i$$

is a solution to $x = y + \lambda Ax$ if and only if y is orthogonal to every eigenvector belonging to λ_ν. The prime on the summation means that eigenvectors belonging to λ_ν are to be omitted, and $\bar{x}$ is *any* linear combination of eigenvectors belonging to λ_ν. Note that the statement that y be orthogonal to every eigenvector belonging to λ_ν is equivalent to the requirement that y be orthogonal to every linearly independent solution of the equation

$$A\xi = \lambda_\nu \xi ,$$

or, since A is self-adjoint,

$$A^\dagger \xi = \lambda_\nu \xi .$$

As in the case of operators of finite rank, we may combine all our results as follows:

Theorem 9.7. The Fredholm alternative holds for any completely continuous Hermitian operator.

This may be compared to Theorem 8.1 and its two corollaries proved for *arbitrary* transformations of finite rank.

Theorem 9.6 also gives us an example of a situation in which the method sketched nonrigorously in Section 7.2 can be carried out to obtain the Green's function of an operator, A. In the terminology of the last chapter, the Green's function of the operator $(I - \lambda A)$ is $(I + \lambda R_A)$; this may be seen by looking at Eq. (9.23). It should also be emphasized that R_A, as defined by Eq. (9.24), is a meromorphic function of λ, with poles located at $\lambda = 1/\lambda_i$.

Now, in most circumstances, when confronted with a linear equation of the type of Eq. (9.17), one will not be fortunate enough to know all the eigenvectors and eigenvalues of A, so that Eqs. (9.21) and (9.22) are often impractical. However, the result of Theorem 8.4 suggests that it may be possible to split A into two parts, $A = A_0 + A_1$, where A_1 is a "small" operator in the sense that $||A_1|| \ll ||A_0||$. It may then happen that one can deal with A_0 in a reasonable fashion. However, before we plunge into actual calculations, it is very important to be certain that the addition of a "small" A_1 to A_0 does not have a severe effect on the spectrum of A_0, since clearly the spectrum of A_0 (and A) will play a very important role in all considerations. It is reassuring to see that if A, A_0, and A_1 are completely continuous, then as $||A_1|| \to 0$, the eigenvalues of A_0 tend toward those of A.

Theorem 9.8. If $A = A_0 + A_1$ and A, A_0, and A_1 are completely continuous Hermitian operators, then we always have

$$\lambda_n^{(0)} - ||A_1|| \leq \lambda_n \leq \lambda_n^{(0)} + ||A_1|| ,$$

where $\lambda_n^{(0)}$ and λ_n are, respectively, the nth eigenvalues of A_0 and A.

Proof. First, consider $\lambda_n^{(0)+}$. Let C be the set of all $y \in H$ such that $||y|| = 1$ and $(y, x_i^{(0)+}) = 0$ for $i = 1, 2, \cdots, n-1$ where $x_i^{(0)+}$ denotes an eigenvector of A_0 with positive eigenvalue $\lambda_i^{(0)+}$. We have

$$\begin{aligned} \sup_{y \in C} (y, Ay) &= \sup_{y \in C} [(y, A_0 y) + (y, A_1 y)] \\ &\leq \sup_{y \in C} (y, A_0 y) + \sup_{y \in C} |(y, A_1 y)| . \end{aligned}$$

But by Theorems 9.1 and 9.5

$$\sup_{y \in C} (y, A_0 y) = \lambda_n^{(0)+} , \qquad \sup_{y \in C} (y, A_1 y) \leq ||A_1|| ,$$

so

$$\sup_{y \in C} (y, Ay) \leq \lambda_n^{(0)+} + ||A_1|| . \tag{9.25}$$

The term on the left of this equation looks very much like λ_n^+, but since y has to be orthogonal to $n-1$ eigenvectors of A_0 rather than $n-1$ eigenvectors of A, it is in fact *not* equal to λ_n^+. However, it is obviously greater than or equal to λ_n^+, since we can clearly find a normalized linear combination of the first n eigenvectors of A which lies in C (i.e., which is orthogonal to the first $n-1$ eigenvectors of A_0). For this vector, η, we will obviously have $(\eta, A\eta) \geq \lambda_n^+$, so

$$\sup_{y \in C} (y, Ay) \geq \lambda_n^+ .$$

Returning to Eq. (9.25), we have

$$\lambda_n^+ \leq \lambda_n^{(0)+} ||A_1|| .$$

For the second half of the inequality, we consider the set D of all y such that $||y|| = 1$ and $(y, x_i^+) = 0$ for $i = 1, 2, \cdots, n-1$. Then we have

$$\begin{aligned} \sup_{y \in D} (y, A_0 y) &= \sup_{y \in D} [(y, Ay) - (y, A_1 y)] \\ &\leq \sup_{y \in D} (y, Ay) + \sup_{y \in D} |(y, A_1 y)| . \end{aligned}$$

Now by the definition of D,

$$\sup_{y \in D} (y, Ay) = \lambda_n^+ , \qquad \text{so} \qquad \sup_{y \in D} (y, A_0 y) \leq \lambda_n^+ + ||A_1|| .$$

But by our previous argument,

$$\sup_{y \in D} (y, A_0 y) \geq \lambda_n^{(0)+} .$$

Hence

$$\lambda_n^{(0)+} \leq \lambda_n^+ + ||A_1|| , \qquad \text{or} \qquad \lambda_n^{(0)+} - ||A_1|| \leq \lambda_n^+ ,$$

which completes the proof for λ_n^+. The case of λ_n^- is handled in an identical fashion.

Now that we have confirmed the stability of the spectrum of a completely continuous Hermitian operator under small perturbations, let us investigate a perturbation approach to the solution of linear equations. We break A into two parts:

$$A = A_0 + A_1 ,$$

such that $||A_1||$ is very small. Then we write Eq. (9.17) as

$$x = y + \lambda A_0 x + \lambda A_1 x ,$$

or more suggestively,

$$(I - \lambda A_0)x = y + \lambda A_1 x .$$

Thus, according to Eq. (9.21), if λ^{-1} is not an eigenvalue of A_0,

$$x = y + \lambda A_1 x + \lambda R_{A_0}(y + \lambda A_1 x) ,$$

or

$$x = y + \lambda R_{A_0} y + \lambda(A_1 + \lambda R_{A_0} A_1)x . \tag{9.26}$$

This presents us with a new linear equation, but now the operator in question is one whose norm is small, since $||A_1||$ is assumed to be small. We have all the time assumed that λ^{-1} is not an eigenvalue of A_0. However, as $||A_1|| \to 0$, we know that the eigenvalues of A_0 get closer and closer to the eigenvalues of A. Thus, in this limit it is equivalent to assume that λ^{-1} is not an eigenvalue of A. As we make $||A_1||$ small, $||A_1 + \lambda R_{A_0} A_1||$ also becomes small; when $||A_1||$ becomes sufficiently small so that $|\lambda| \cdot ||A_1 + \lambda R_{A_0} A_1|| < 1$, then we can solve Eq. (9.26) by the Neumann series. Thus, in the limit $||A_1|| \to 0$,

$$x \to y + \lambda R_{A_0} y ,$$

if λ is not an eigenvalue of A. From Theorem 8.4, we know that it is always possible to approximate A in operator norm by a transformation of finite rank. Thus we can reduce the original problem to a matrix problem without knowing the eigenfunctions of A, by making use of the results of Section 8.6, in particular, Eq. (8.88). If A is an integral operator, the method of approximation will involve replacing the kernel by a series of step-functions, a method which we will discuss in more detail in the next section.

Now let us move on to the eigenvalue problem. For convenience, we write A in the form

$$A = A_0 + \epsilon A_1 ,$$

where we have put a constant, ϵ, in front of A_1 to emphasize the fact that this term will be small in norm. A, A_0, and A_1 are all completely continuous,

Hermitian operators. Consider the eigenvalue problem

$$Ax_n \equiv (A_0 + \epsilon A_1)x_n = \lambda_n x_n \,, \tag{9.27}$$

or, assuming that $\lambda_n \neq 0$,

$$\left(I - \frac{1}{\lambda_n} A_0\right) x_n = \frac{1}{\lambda_n} \epsilon A_1 x_n \,.$$

By our previous work, this linear equation with inhomogeneous term equal to $\epsilon A_1 x_n/\lambda_n$ can be solved by writing

$$x_n = \left(I - \frac{1}{\lambda_n} A_0\right)^{-1} \frac{1}{\lambda_n} \epsilon A_1 x_n \,.$$

By Eqs. (9.23) and (9.24), this is just

$$\begin{aligned} x_n &= \sum_{m=1}^{\infty} \left(1 - \frac{\lambda_m^{(0)}}{\lambda_n}\right)^{-1} \frac{1}{\lambda_n} \epsilon (x_m^{(0)}, A_1 x_n) x_m^{(0)} \\ &= \epsilon \sum_{m=1}^{\infty} (\lambda_n - \lambda_m^{(0)})^{-1} (x_m^{(0)}, A_1 x_n) x_m^{(0)} \,, \end{aligned} \tag{9.28}$$

where the $x_m^{(0)}$ are the eigenvectors of A_0 and $\lambda_m^{(0)}$ are the corresponding eigenvalues. We assume that λ_n is not equal to any of the eigenvalues of A_0. When $\epsilon ||A_1||$ is sufficiently small, this certainly will be the case, *except* for the eigenvalue $\lambda_n^{(0)}$, which tends toward λ_n as $\epsilon \to 0$. We also assume for convenience that the spectrum of A_0 is nondegenerate. With the above difficulty in mind, let us isolate the possibly divergent denominator in Eq. (9.28) by writing

$$x_n = \frac{(x_n^{(0)}, \epsilon A_1 x_n)}{\lambda_n - \lambda_n^{(0)}} x_n^{(0)} + \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n)}{\lambda_n - \lambda_m^{(0)}} x_m^{(0)} \,, \tag{9.29}$$

where the prime on the summation means that the term with $m = n$ is omitted. Now Eq. (9.29) looks very much like an ordinary inhomogeneous linear equation. Before attempting to solve it by the methods we have developed, we should say something about the suspicious-looking constant multiplying $x_n^{(0)}$ in Eq. (9.29). Taking the inner product on both sides of Eq. (9.27) by $x_n^{(0)}$ and using the fact that

$$A_0 x_n^{(0)} = \lambda_n^{(0)} x_n^{(0)} \,,$$

we get

$$(x_n^{(0)}, \epsilon A_1 x_n) = (\lambda_n - \lambda_n^{(0)})(x_n^{(0)}, x_n) \,. \tag{9.30}$$

This tells us that as $\lambda_n^{(0)} \to \lambda_n$, $(x_n^{(0)}, \epsilon A_1 x_n) \to 0$ in such a way that

$$\frac{(x_n^{(0)}, \epsilon A_1 x_n)}{\lambda_n - \lambda_n^{(0)}} = (x_n^{(0)}, x_n) < \infty \,.$$

This determines the constant, and allows us to write Eq. (9.29) as

$$x_n = (x_n^{(0)}, x_n) x_n^{(0)} + \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n)}{\lambda_n - \lambda_m^{(0)}} x_m^{(0)} \,. \tag{9.31}$$

Only under rather unusual circumstances would $(x_n^{(0)}, x_n) = 0$, and certainly not if ϵ is sufficiently small, for under this assumption we can generate a contradiction. Assuming that $(x_n^{(0)}, x_n) = 0$, Eq. (9.31) becomes

$$x_n = \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n)}{\lambda_n - \lambda_m^{(0)}} x_m^{(0)} .$$

It follows that

$$||x_n|| = \epsilon \left[\sum_{m=1}^{\infty}{}' \frac{|(x_m^{(0)}, A_1 x_n)|^2}{(\lambda_n - \lambda_m^{(0)})^2} \right]^{1/2} \leq \frac{\epsilon}{\Delta_n} \left[\sum_{m=1}^{\infty}{}' |(x_m^{(0)}, A_1 x_n)|^2 \right]^{1/2} ,$$

where Δ_n is the minimum of $|\lambda_n - \lambda_m^{(0)}|$ taken over all eigenvalues of A_0 except $\lambda_n^{(0)}$. For ϵ sufficiently small, Δ_n tends toward a number which is independent of ϵ. Thus, by completeness, we have

$$||x_n|| \leq \frac{\epsilon}{\Delta_n} (A_1 x_n, A_1 x_n)^{1/2} = \frac{\epsilon}{\Delta_n} ||A_1 x_n|| ,$$

so that

$$||x_n|| \leq \frac{\epsilon ||A_1||}{\Delta_n} ||x_n|| \qquad \text{or} \qquad 1 \leq \frac{\epsilon ||A_1||}{\Delta_n} .$$

For ϵ sufficiently small, we clearly have a contradiction.

Since the equation for x_n is homogeneous, x_n is arbitrary up to a multiplicative constant; setting $(x_n^{(0)}, x_n) = 1$ is a convenient way to fix that constant. With this choice, Eq. (9.31) takes the simple form

$$x_n = x_n^{(0)} + \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n)}{\lambda_n - \lambda_m^{(0)}} x_m^{(0)} . \tag{9.32}$$

Also, according to Eq. (9.30),

$$\lambda_n = \lambda_n^{(0)} + \epsilon (x_n^{(0)}, A_1 x_n) . \tag{9.33}$$

Equations. (9.32) and (9.33) are the starting points for studying perturbations in the spectra of completely continuous Hermitian operators.

Certainly the most promising attack on these equations for the case of $\epsilon ||A_1||$ small but finite would seem to be through iterative techniques. However, we shall have to proceed in a slightly different manner than in Sections 8.1 and 8.3, since the operator on the right-hand side of Eq. (9.32) contains an unknown quantity, λ_n. On the other hand, Eq. (9.33) is an equation for λ_n, but in terms of the unknown vector x_n. Thus we have a pair of coupled nonlinear equations for x_n and λ_n. Now we have seen that the norm of the operator on the right-hand side of Eq. (9.32) is equal to or less than $\epsilon ||A_1||/\Delta_n$. Therefore, if this quantity is less than 1, we can iterate Eq. (9.32) to obtain

$$x_n = x_n^{(0)} + \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{\lambda_n - \lambda_m^{(0)}} x_m^{(0)}$$
$$+ \epsilon^2 \sum_{l=1}^{\infty}{}' \sum_{m=1}^{\infty} \frac{(x_m^{(0)}, A_1 x_l^{(0)})(x_l^{(0)}, A_1 x_n^{(0)})}{(\lambda_n - \lambda_m^{(0)})(\lambda_n - \lambda_l^{(0)})} x_m^{(0)} + \cdots . \tag{9.34}$$

When we have computed x_n to sufficient accuracy, we insert Eq. (9.34) into Eq. (9.33) and thus obtain a relation for λ_n. Of course, in general, this equation will be extremely complicated.

Another possibility would be to substitute Eq. (9.33) into Eq. (9.34) to obtain

$$x_n = x_n^{(0)} + \epsilon \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n^{(0)}) x_m^{(0)}}{\lambda_n^{(0)} - \lambda_m^{(0)} + \epsilon(x_n^{(0)}, A_1 x_n)}$$
$$+ \epsilon^2 \sum_{m=1}^{\infty}{}' \sum_{l=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_l^{(0)})(x_l^{(0)}, A_1 x_n^{(0)})}{[\lambda_n^{(0)} - \lambda_m^{(0)} + \epsilon(x_n^{(0)}, A_1 x_n)][\lambda_n^{(0)} - \lambda_l^{(0)} + \epsilon(x_n^{(0)}, A_1 x_n)]} x_m^{(0)}$$
$$+ \cdots . \tag{9.35}$$

If $\epsilon ||A_1||$ is small enough that

$$\left| \frac{\epsilon(x_n^{(0)}, A_1 x_n)}{\lambda_n^{(0)} - \lambda_m^{(0)}} \right| < 1 , \tag{9.36}$$

for all $m \neq n$, then we can expand each of the denominators occurring in Eq. (9.35) in a power series in ϵ:

$$[\lambda_n^{(0)} - \lambda_m^{(0)} + \epsilon(x_n^{(0)}, A_1 x_n)]^{-1} = (\lambda_n^{(0)} - \lambda_m^{(0)})^{-1} \sum_{\mu=0}^{\infty} (-1)^\mu \epsilon^\mu \frac{(x_n^{(0)}, A_1 x_n)^\mu}{(\lambda_n^{(0)} - \lambda_m^{(0)})^\mu}. \tag{9.37}$$

Putting this result into Eq. (9.35), we obtain for the first few terms of the combined iteration and denominator expansion

$$x_n = x_n^{(0)} + \sum_{m=1}^{\infty}{}' \sum_{\mu=0}^{\infty} (-1)^\mu \epsilon^{\mu+1} \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{[\lambda_n^{(0)} - \lambda_m^{(0)}]^{\mu+1}} (x_n^{(0)}, A_1 x_n)^\mu x_m^{(0)}$$
$$+ \sum_{m=1}^{\infty}{}' \sum_{l=1}^{\infty}{}' \sum_{\mu=0}^{\infty} \sum_{\nu=0}^{\infty} (-1)^{\mu+\nu} \epsilon^{\mu+\nu+2}$$
$$\times \frac{(x_m^{(0)}, A_1 x_l^{(0)})(x_l^{(0)}, A_1 x_n^{(0)})}{[\lambda_n^{(0)} - \lambda_m^{(0)}]^{\mu+1}[\lambda_n^{(0)} - \lambda_l^{(0)}]^{\nu+1}} (x_n^{(0)}, A_1 x_n)^{\mu+\nu} x_m^{(0)}$$
$$+ \cdots . \tag{9.38}$$

If ϵ is small enough, we obtain a convergent power series expansion for x_n. If we now write

$$x_n = \sum_{l=0}^{\infty} \epsilon^l f_n^{(l)} \tag{9.39}$$

and insert this expression into both sides of (9.38), we can calculate the vectors $f_n^{(l)}$ by equating the coefficients of equal powers of ϵ on either side of the equation. Doing this, we find for the first few $f_n^{(l)}$

$$f_n^{(0)} = x_n^{(0)} ,$$

$$f_n^{(1)} = \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{\lambda_n^{(0)} - \lambda_m^{(0)}} x_m^{(0)} ,$$

and

$$f_n^{(2)} = \sum_{m=1}^{\infty}{}' \sum_{l=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_l^{(0)})(x_l^{(0)}, A_1 x_n^{(0)})}{(\lambda_n^{(0)} - \lambda_m^{(0)})(\lambda_n^{(0)} - \lambda_l^{(0)})} x_m^{(0)}$$

$$- (x_n^{(0)}, A_1 x_n^{(0)}) \sum_{m=1}^{\infty}{}' \frac{(x_m^{(0)}, A_1 x_n^{(0)})}{(\lambda_n^{(0)} - \lambda_m^{(0)})^2} x_m^{(0)} . \tag{9.40}$$

Putting these into Eq. (9.39), we obtain an expression for x_n through order ϵ^2. With this result, we then use Eq. (9.33) to calculate λ_n through third order in ϵ. The eigenvalue λ_n can also be written in a power series, the first term of which is $\lambda_n^{(0)}$. We have

$$\lambda_n = \sum_{l=0}^{\infty} \epsilon^l \lambda_n^{(l)} . \tag{9.41}$$

Using x_n as given by Eqs. (9.39) and (9.40), we find that

$$\lambda_n^{(1)} = (x_n^{(0)}, A_1 x_n^{(0)}), \qquad \lambda_n^{(2)} = \sum_{m=1}^{\infty}{}' \frac{|(x_m^{(0)}, A_1 x_n^{(0)})|^2}{\lambda_n^{(0)} - \lambda_m^{(0)}} \tag{9.42}$$

and similarly for higher-order terms.

These results agree exactly with those obtained in Rayleigh-Schrödinger perturbation theory (Section 4.11). Thus we have been able to give a derivation of Rayleigh-Schrödinger perturbation theory, at least in the nondegenerate case, and have shown that if ϵ is sufficiently small, the perturbation series for x_n will converge. It is difficult to give a more precise statement of convergence requirements, but it is fairly clear from the above discussion that if the difference between $\lambda_n^{(0)}$ and the nearest eigenvalue of A_0 is much greater in magnitude than $(\lambda_n - \lambda_n^{(0)})$, then the method has a good chance of converging. As a rule of thumb, this estimate is often helpful.

The results of Eqs. (9.39), (9.40), (9.41), and (9.42) suffer from the drawback that they require the knowlege of *all* the eigenvectors and eigenvalues of A_0. There are several ways to avoid this difficulty. One way is to make use of the proved *existence* of series expansions for λ_n and x_n and then obtain coupled equations for the vectors $f_n^{(l)}$ involving the $\lambda_n^{(l)}$ as was done in Section 4.11. In this way one obtains, for example,

$$\begin{aligned}
(A_0 - \lambda_n^{(0)}) f_n^{(1)} &= -(A_1 - \lambda_n^{(1)}) x_n^{(0)} , \\
(A_0 - \lambda_n^{(0)}) f_n^{(2)} &= -(A_1 - \lambda_n^{(1)}) f_n^{(1)} + \lambda_n^{(2)} x_n^{(0)} , \\
(A_0 - \lambda_n^{(0)}) f_n^{(3)} &= -(A_1 - \lambda_n^{(1)}) f_n^{(2)} + \lambda_n^{(2)} f_n^{(1)} + \lambda_n^{(3)} x_n^{(0)} ,
\end{aligned} \tag{9.43}$$

and so forth, just as in Section 4.11. These equations involve only $x_n^{(0)}$ and $\lambda_n^{(0)}$ as their starting point, and one has just a system of inhomogeneous equations to solve. This system was discussed in some detail in Section 4.11.

Another approach is to make a direct attack on A itself, using the maximum and minimum principles of Theorem 9.5. Since we know that for any $y \in H$, $||y|| = 1$,

$$(y, Ay) \leq \lambda_1^+ ,$$

we can build "trial vectors," y_t, containing many variable parameters and then maximize (y, Ay) with respect to these parameters, subject to the constraint $||y_t|| = 1$. If the parameters occur linearly in y_t it can be shown (see Problem 5) that the problem reduces to a matrix problem, with the linear parameters in y_t being the unknowns. As we perform this process with more and more parameters, the successive values of (y_t, Ay_t) must increase toward, but never exceed, λ_1^+. If one chooses trial functions sensibly, this process can very quickly yield excellent approximations for λ_1^+ and x_1^+ (the function y_t which gives the good approximation to λ_1^+).

As an example of this method, we consider the equation

$$Ax = \int_0^1 t_<(1 - t_>)x(t')\,dt' = \lambda x(t) ,$$

where A is defined on the space of square-integrable function on $[0, 1]$. We already know (see Section 7.1) that this equation has the eigenfunctions

$$x_n(t) = \sqrt{2}\sin(n\pi t)$$

and eigenvalues

$$\lambda_n = 1/n^2\pi^2$$

for the boundary conditions $x_n(0) = 0 = x_n(1)$. Now the simplest normalized trial function which vanishes at $t = 0$ and $t = 1$ is

$$y_t(t) = \sqrt{30}\,t(1 - t) .$$

If we calculate (y_t, Ay_t), we find that

$$(y_t, Ay_t) = 17/168 = 0.10119 ,$$

which is less than $1/\pi^2 = 0.10132$, but not by much! Thus by making the most obvious choice, we get a result which is extremely close to the correct one. Needless to say, in most problems one is not quite so lucky. One can also calculate $||x_1^+ - y_t||$:

$$\begin{aligned} ||x_1^+ - y_t|| &= ||\sqrt{2}\sin(\pi t) - \sqrt{30}\,t(1 - t)|| \\ &= [2(1 - 4\sqrt{60}/\pi^3)]^{1/2} = 0.038 . \end{aligned}$$

Thus y_t approximates x_1^+ very closely in norm. If one wanted still greater accuracy, one would choose a trial function with several parameters, for example,

$$y_t = at(1 - t) + bt^2(1 - t)^2 ,$$

and then maximize (y_t, Ay_t) with respect to a and b subject to $||y_t|| = 1$. In practice, it is convenient to handle the constraint on the norm of y_t by using a Lagrange multiplier. Clearly, $\lambda_2^+, \lambda_3^+, \cdots$ can be computed in the same way by using more constraint equations to ensure that the proper orthogonality properties are satisfied.

9.3 FINITE-RANK TECHNIQUES FOR EIGENVALUE PROBLEMS

For most realistic physical problems, the chances of obtaining closed-form solutions are rather remote. This perhaps comes as a shock to the student who has been exposed to the beautiful exact solutions which can be obtained for the harmonic oscillator and the $1/r$ potential in both classical and quantum mechanics. However, problems like these are the exception rather than the rule in physics (although very important exceptions, since they often provide the insight necessary for the understanding of more complicated problems). In the case of the Coulomb potential, for example, all one need do is to consider the potential as arising from a charge distribution of finite extent (a proton) rather than a point charge, and one is immediately forced to look beyond exact solutions to some sort of approximation procedure.

However, in the problem just mentioned, the effect of the finite proton size can be thought of as a small perturbation so that the techniques discussed in Section 9.2 can be successfully applied. The type of problem with which we shall be concerned in this section is one in which the idea of a small perturbation of a well-understood physical system is inadequate. For such problems, we will make use of a different sort of approximation, namely, the notion of approximation by an operator of finite rank, which was discussed in Section 8.7.

What we want to do is to consider the eigenvalue problem of Eq. (9.27) in the case when $\epsilon||A_1||$ tends to zero, i.e., when we approximate A extremely closely by A_0, and in principle, arbitrarily closely (see Theorem 8.4). In this case, according to our work in the last section

$$x_n^{(0)} \to x_n$$

and

$$\lambda_n^{(0)} \to \lambda_n .$$

Let us focus our attention on a Hermitian operator whose kernel is $a(t, t')$ with t and t' in an interval $[a, b]$. We have already encountered such problems in Chapter 7 when discussing Schrödinger's integral equation. The basic idea which we will use is to divide $[a, b]$ into N small segments, each of length $(b - a)/N$, and define the set of orthogonal functions

$$\phi_n^{(N)}(t) = \begin{cases} 1 , & a + \dfrac{n-1}{N} l \leq t \leq a + \dfrac{n}{N} l , \\ 0 , & \text{otherwise} , \end{cases}$$

where $l = b - a$ and $n = 1, 2, \cdots, N$. If we wished, we could normalize $\phi_n^{(N)}(t)$ to 1 by multiplying it by $N^{1/2}$. Now we can approximate any kernel in L_2 arbitrarily closely by such functions; if the kernel is defined on an infinite range, then we truncate it by setting it equal to zero when t or t' is outside some large, but finite interval. It is usually possible to determine a sensible point of truncation by inspection of the kernel. With the $\phi_n^{(N)}$, we can approximate $a(t, t')$ arbitrarily closely by some $a_N(t, t')$, satisfying

$$||a(t, t') - a_N(t, t')|| < \epsilon$$

for any preassigned ϵ. If we call the operator associated with $a_N(t, t')$, A_N, then

$$||A - A_N|| < \epsilon .$$

Thus according to Theorem 9.8 the eigenvalues of A_N tend toward those of A, and according to Eq. (9.32) the eigenvectors of A_N tend in norm toward those of A. Hence the problem reduces to solving the equation

$$A_N x_n^{(0)}(t) = \lambda_n^{(0)} x_n^{(0)}(t) , \tag{9.44}$$

for some sufficiently large N. In terms of the $\phi_n^{(N)}$, the kernel of A_N can be written as

$$a_N(t, t') = \sum_{\mu,\nu=1}^{N} M_{\mu\nu}\phi_\mu^{(N)}(t)\phi_\nu^{(N)}(t') , \tag{9.45}$$

where

$$M_{\mu\nu} = (N/l)^2 \int dt \int dt' \; \phi_\mu^{(N)}(t)\phi_\nu^{(N)}(t')a(t, t') . \tag{9.46}$$

For convenience, we take $a(t, t')$ to be real. Note that $M_{\mu\nu}$ is simply the *average value* of the function $a(t, t')$ on the $\mu\nu$-square of the two-dimensional grid into which we have divided the tt'-plane. Using Eq. (9.45), we get for Eq. (9.44)

$$\sum_{\mu,\nu=1}^{N} \phi_\mu^{(N)}(t) M_{\mu\nu} \int \phi_\nu^{(N)}(t') x_n^{(0)}(t') \, dt' = \lambda_n^{(0)} x_n^{(0)}(t) .$$

If we write

$$C_\nu^{(n)} \equiv (N/l) \int \phi_\nu^{(N)}(t') x_n^{(0)}(t') \, dt'$$

for the average value of $x_n^{(0)}$ on the νth subinterval of $[a, b]$, then our equation becomes

$$\sum_{\mu,\nu=1}^{N} \phi_\mu^{(N)}(t) M_{\mu\nu} C_\nu^{(n)} \frac{l}{N} = \lambda_n^{(0)} x_n^{(0)}(t) . \tag{9.47}$$

Thus all we need to do is evaluate the scalars $C_\nu^{(n)}$ by using Eq. (9.47). Multiplying both sides by $\phi_{\nu'}^{(N)}(t)$ and integrating, we get

$$\frac{l}{N} \sum_{\nu=1}^{N} M_{\nu'\nu} C_\nu^{(n)} = \lambda_n^{(0)} C_{\nu'}^{(n)} \tag{9.48}$$

for $\nu' = 1, 2, \cdots, N$. Equation (9.48) thus reduces our problem to one of finding the eigenvalues and eigenvectors of a finite-dimensional matrix, M. With Eq. (9.48) solved for $\lambda_n^{(0)}$ and $C_\nu^{(n)}$ ($\nu = 1, 2, \cdots, N$), then Eq. (9.47) tells us that

$$x_n^{(0)}(t) = \sum_{\nu=1}^{N} C_\nu^{(n)} \phi_\nu^{(N)}(t) . \tag{9.49}$$